wellcome connecting science

Hybrid course

CRISPR Screening and Analysis

Wellcome Genome Campus, UK

Online learning dates: 28 July - 1 August | In-person training dates: 21-26 September

This two-phase course blends online learning with in-person, hands-on training to help you learn how to apply CRISPR screening and analysis techniques to real-world scenarios.

Experts from the Wellcome Sanger Institute and Open Targets will teach you everything you need to know about pooled and arrayed CRISPR screens.

The course will focus on three therapeutic areas where pooled and arrayed screens are commonly used:

- **Oncology**
- **Neurodegeneration**
- **Immunology**

Apply by 3 June to be considered for a place!

SCAN ME

CRISPR - One of the biggest scientific stories of the century

Gene Editing workflows and integrated concepts

Kalpana Surendranath **MSc PhD PGCHE SFHEA FRSB**
Associate professor in genome engineering
University of Westminster
Director - Gene Editors of the Future
www.geneeditorsofthefuture.uk

UNIVERSITY OF WESTMINSTER

GENE EDITORS
OF THE FUTURE

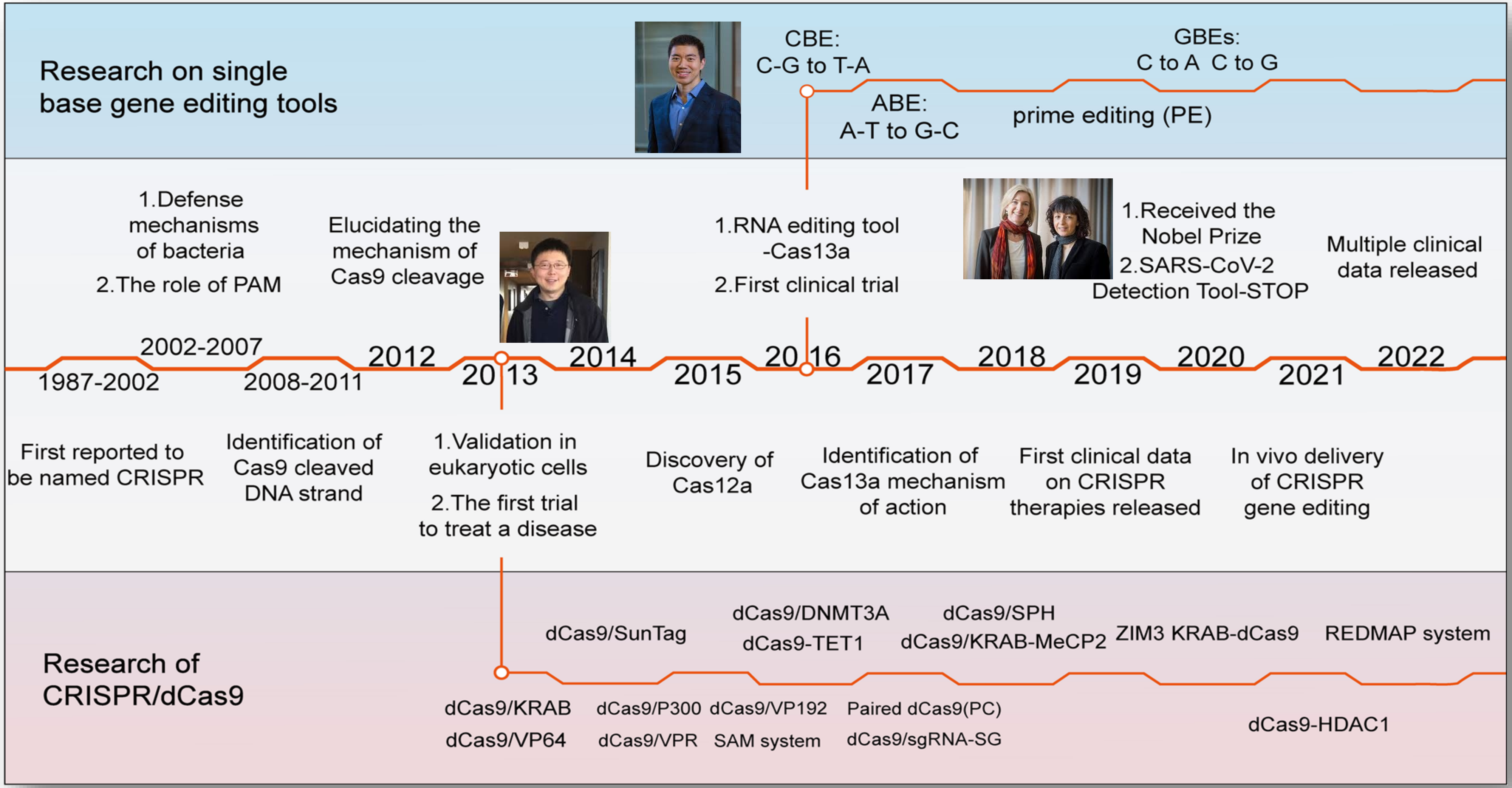
Genome
Engineering
Laboratory

Learning objectives

At the end of the session, you are expected have an appreciation of:

- The origins
- Fundamentals of **CRISPR** /Cas9 tools
- Workflow of **CRISPR** in generating *cellular models of* disease- Guide Design and Clonal isolation
- Benchling and vendor tools

CRISPR Timeline

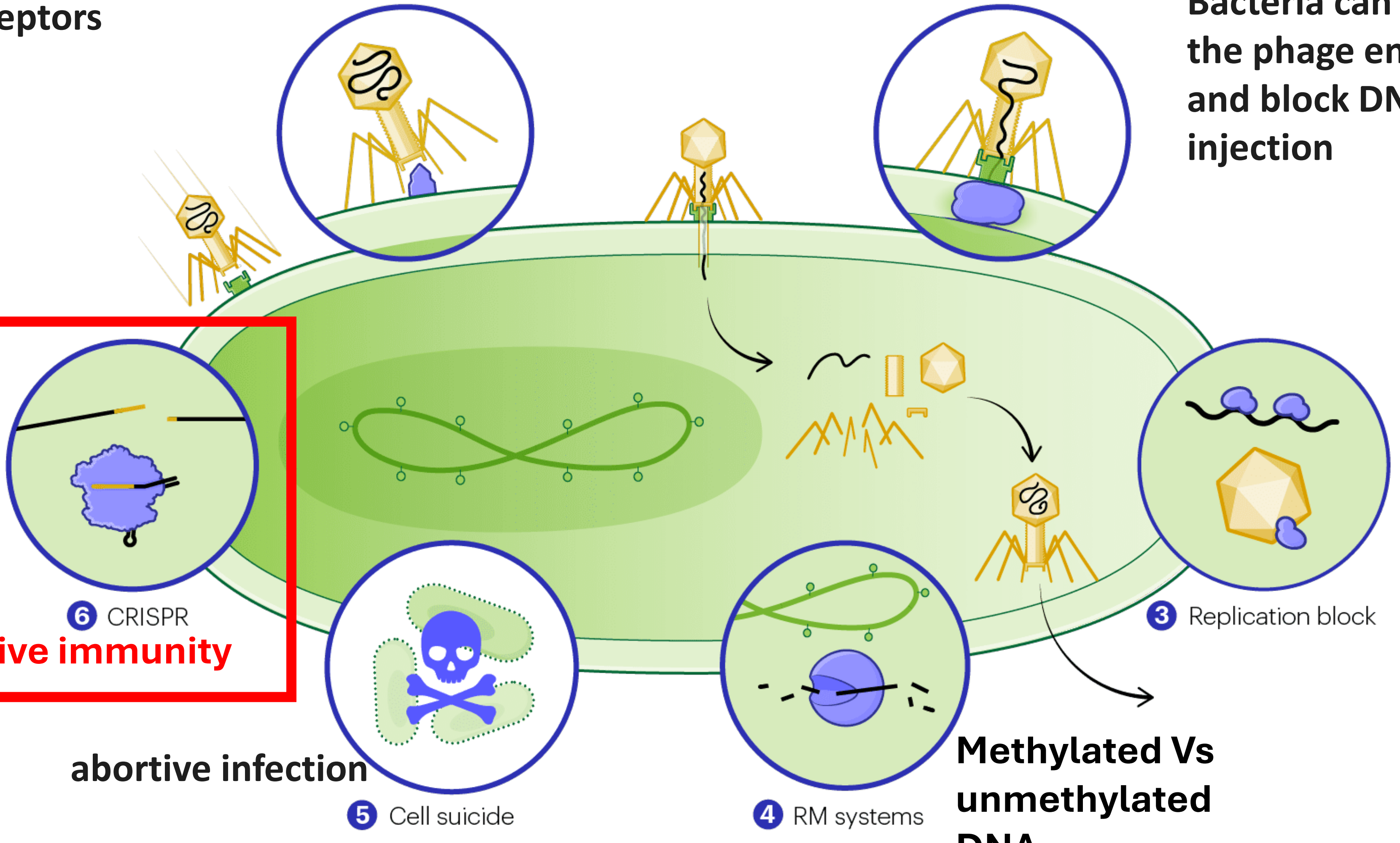


https://www.nature.com/articles/s41392-023-01309-7

The molecular arms race

Altering the structure of their receptors

Bacteria can clog the phage entryway and block DNA injection

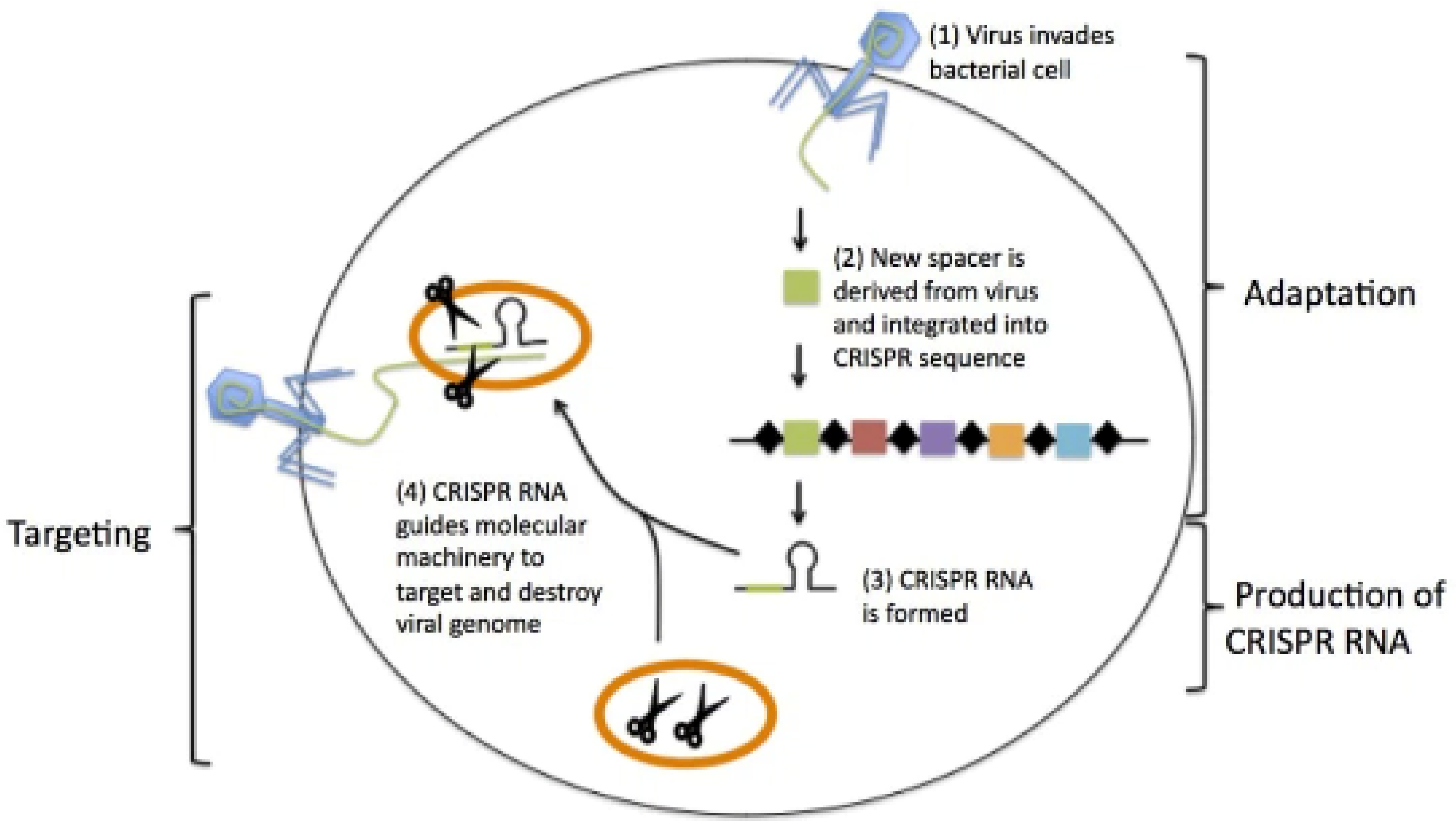


Adaptive immunity

abortive infection

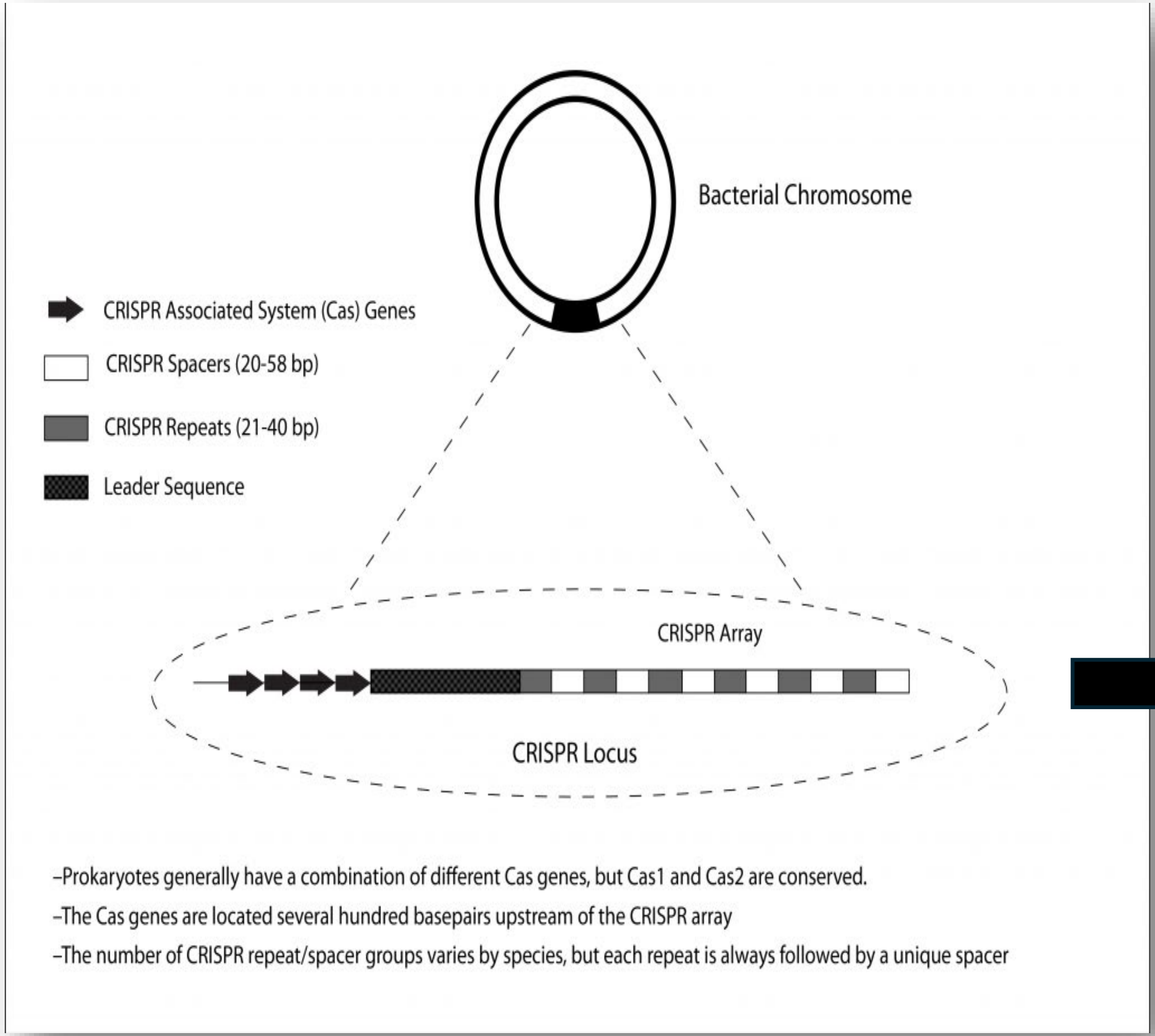
Methylated Vs unmethylated DNA

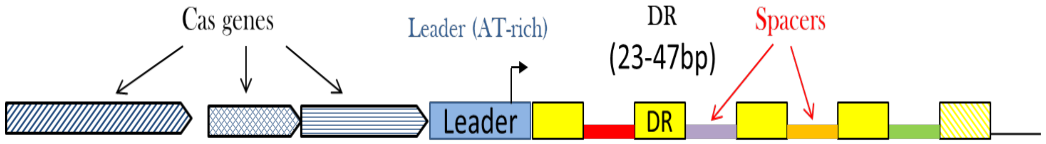
Bacterial adaptive immunity



CRISPR-conferred phage resistance relies on incorporation of short phage-derived sequences within specific loci of the bacterial genome. In effect, the bacterial cell becomes immune to phages which carry homologous sequences

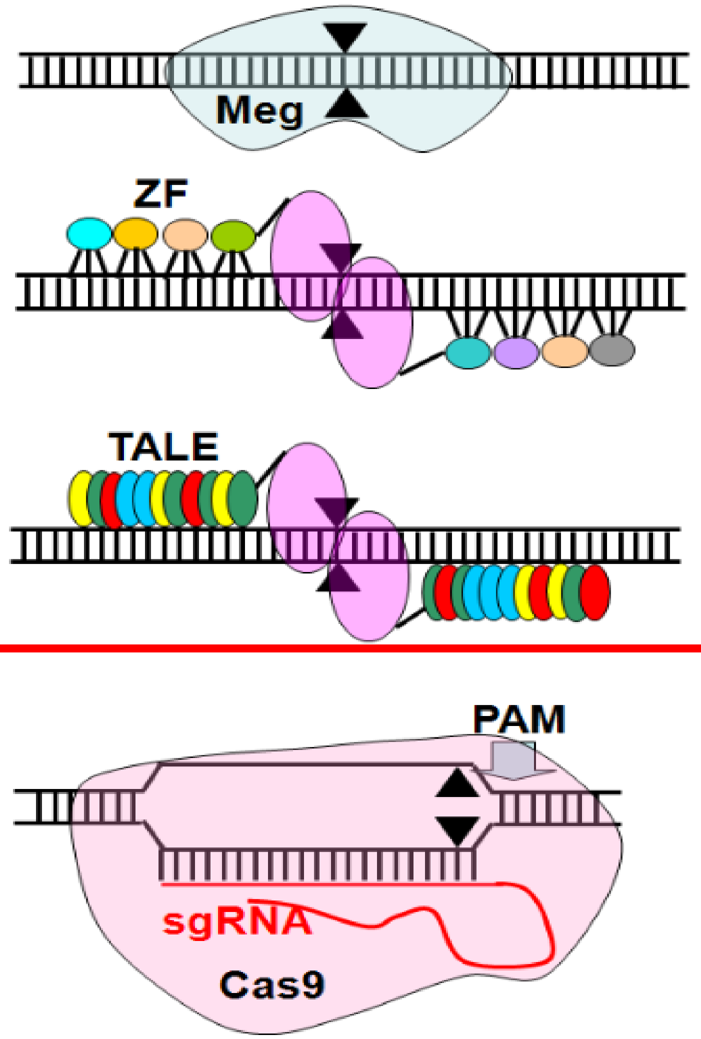
CRISPR Locus





TATAAATCAGTAAGTTACGAGGCCTCGAAAAAAGAGGGTTTCTGGCAGGAAAAACTCGGTATTTCCTTTT
CCTTCAAATGGTTATAGGTTTTCAGGGCTAGTTCACTGCCGTATAGGCAGCTAAGAAA GCAGCGATCAAC
GGCCATATCTACGAGCTGGAGTTCACTGCCGTATAGGCAGCTAAGAAACTGACCCGCCGCATGGTGCTGG
GCCTGCAGGA GTTCACTGCCGTATAGGCAGCTAAGAAATGCCAGCGGGCGGTATGCGCCTGCGGAGCT
TC GTTCACTGCCGTATAGGCAGCTAAGAAATCGATCAGCTTCGCGGCGCGGGCGGTGTATTCTTTCACTG
CCACATAGGTCGTCAAGAAACGGCCAGCCTGCGCCGTTCATTGCCGTGTAGTCCCGTAGGGCGAATGCCG

Programmable Nucleases



Meganuclease — 20-40 bp/Enzyme

ZFN — 3 bp/Finger

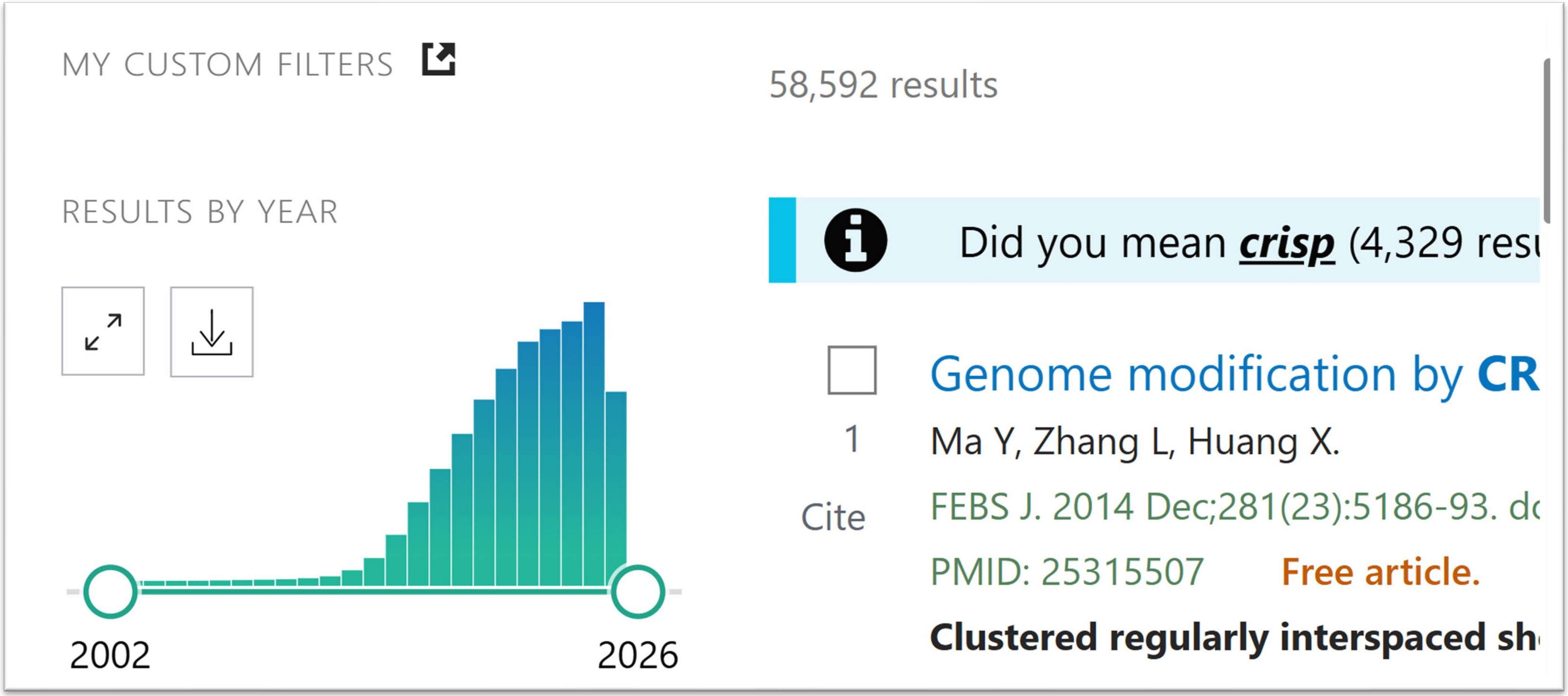
TALEN — 1 bp/Module

CRISPR-Cas — 1 bp/Base

High cost

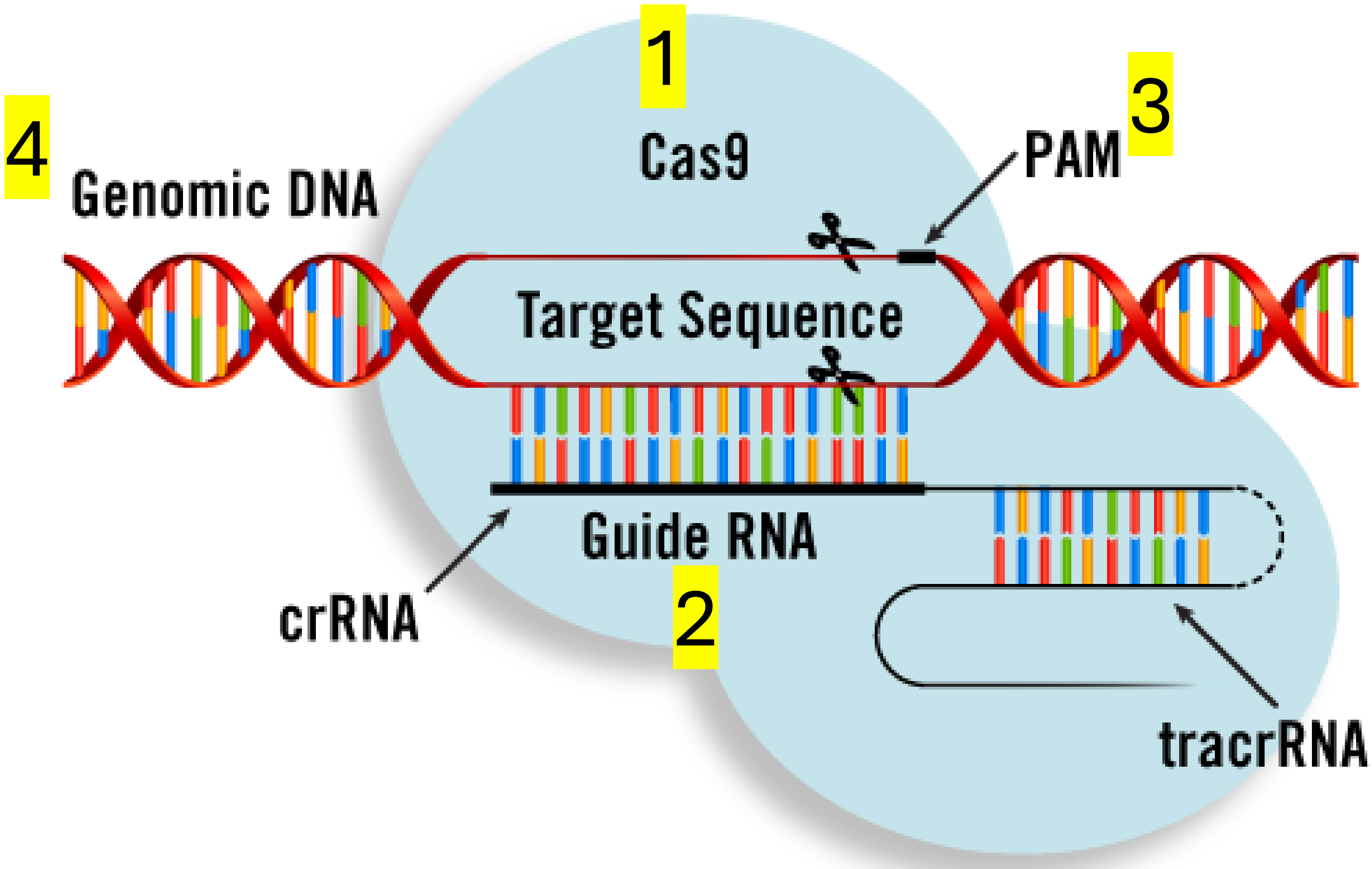
difficult

As of July 28, 2025

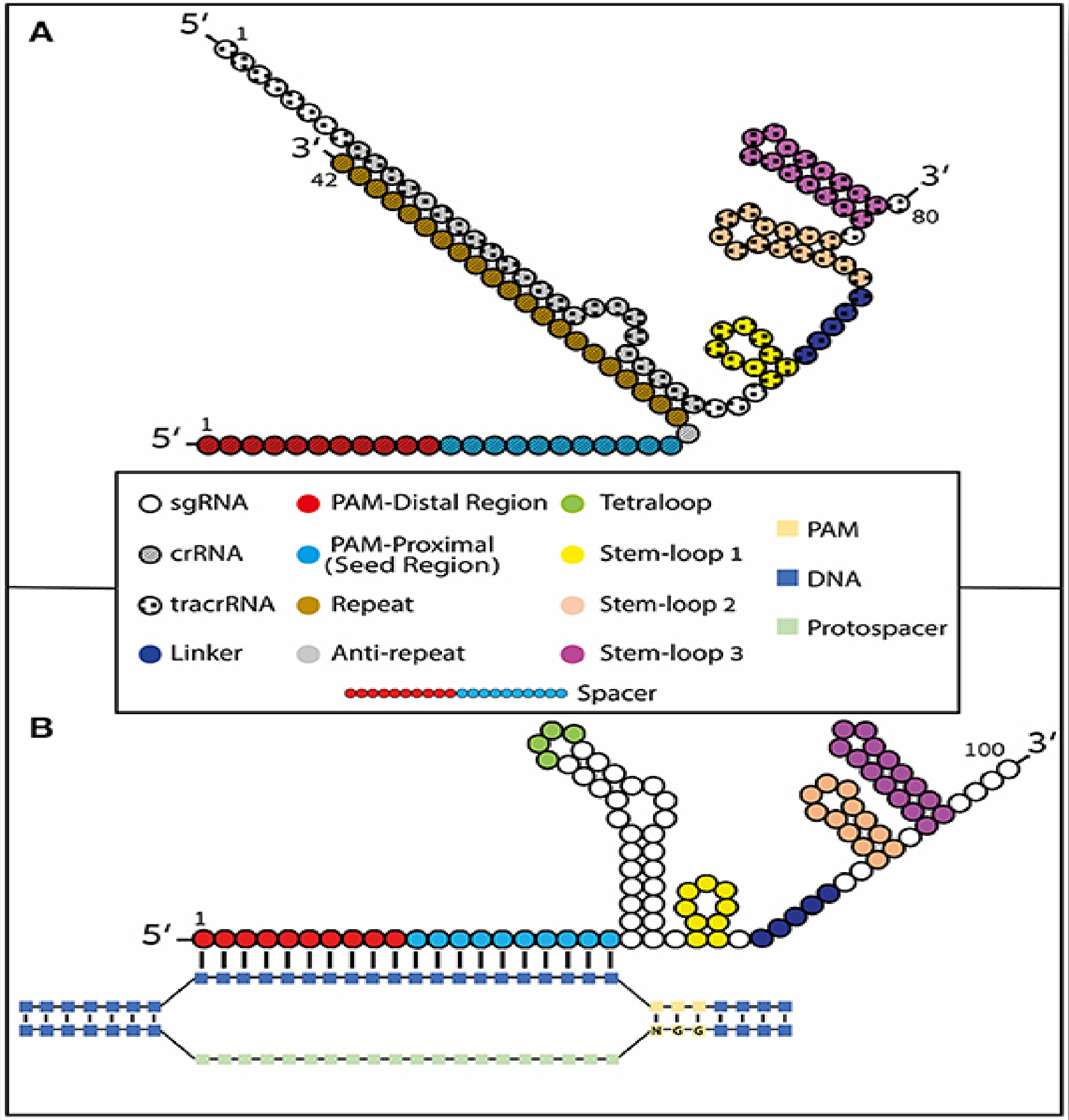


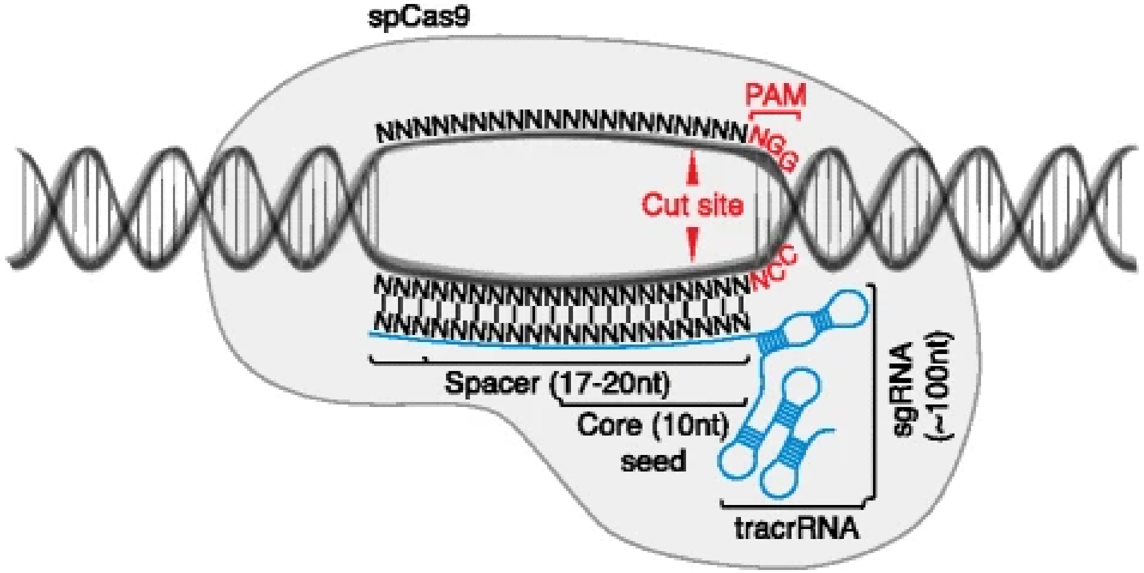
https://pubmed.ncbi.nlm.nih.gov/?term=CRISPR

CRISPR-Cas complex basics



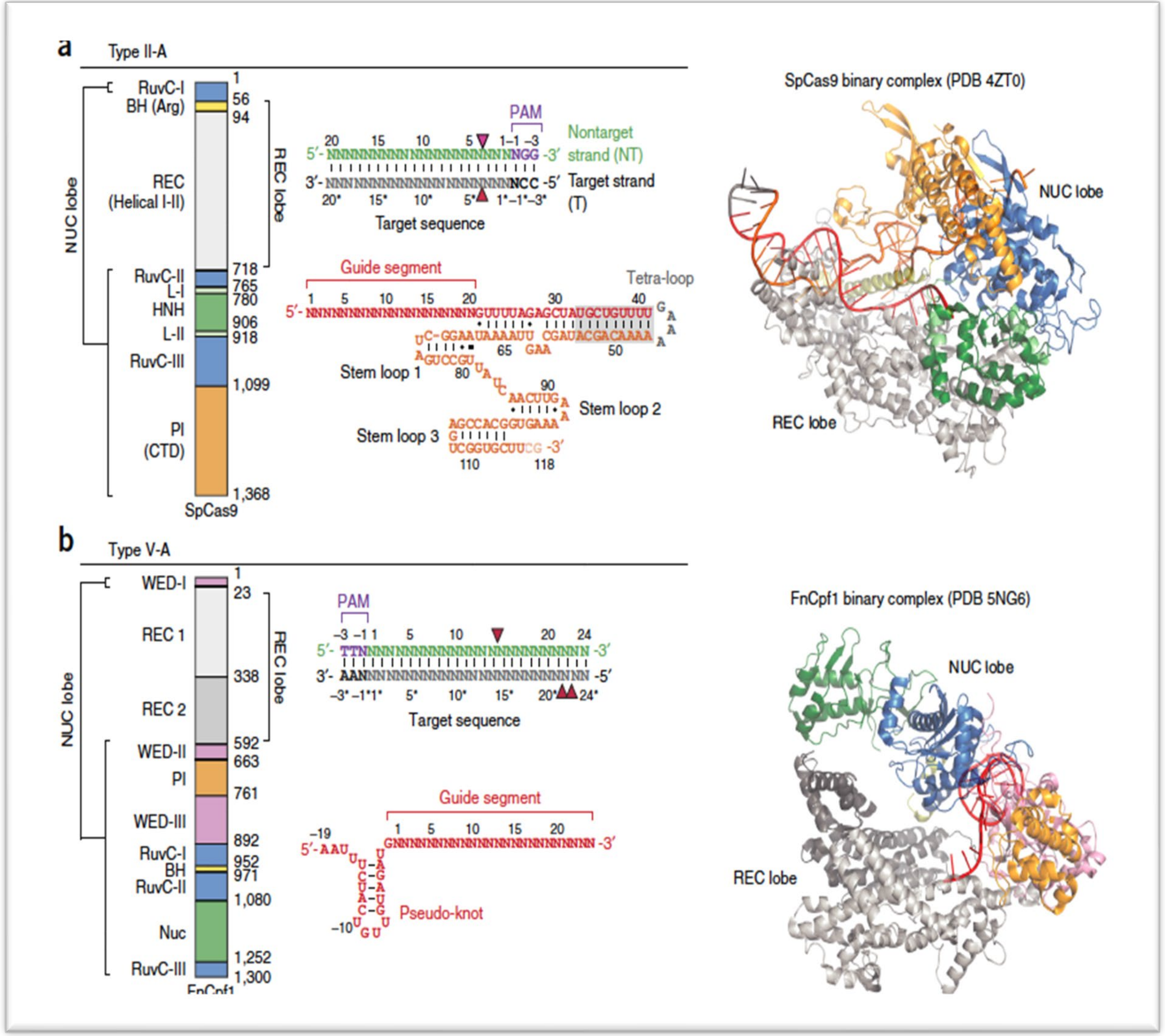
Guide RNAs





https://www.frontiersin.org/journals/genome-editing/articles/10.3389/fgeed.2020.617910/full

CRISPR /Cas9 complex



Nishimasu H, Ran FA, Hsu PD, et al. Crystal structure of Cas9 in complex with guide RNA and target DNA. *Cell*. 2014;156(5):935-949

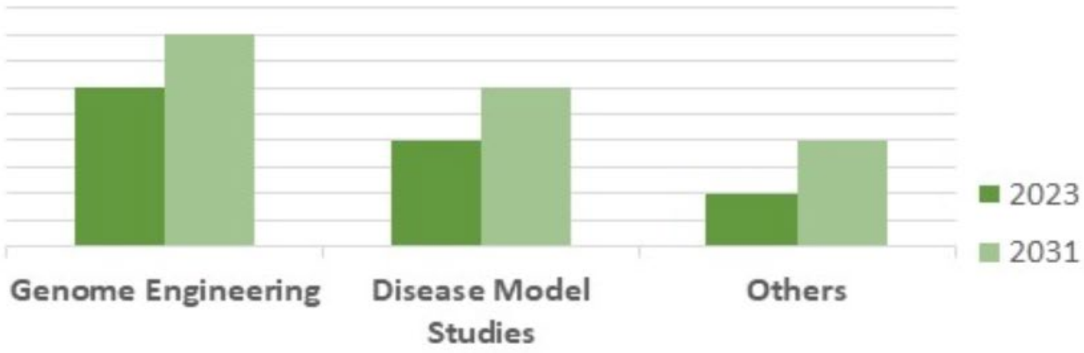
Guide RNAs

By gRNA Type :

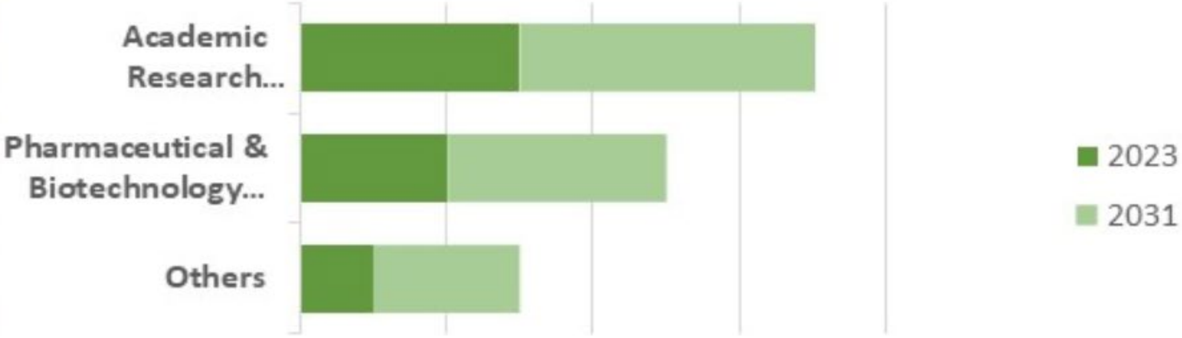
2023

2031

By Application :



By End-User:



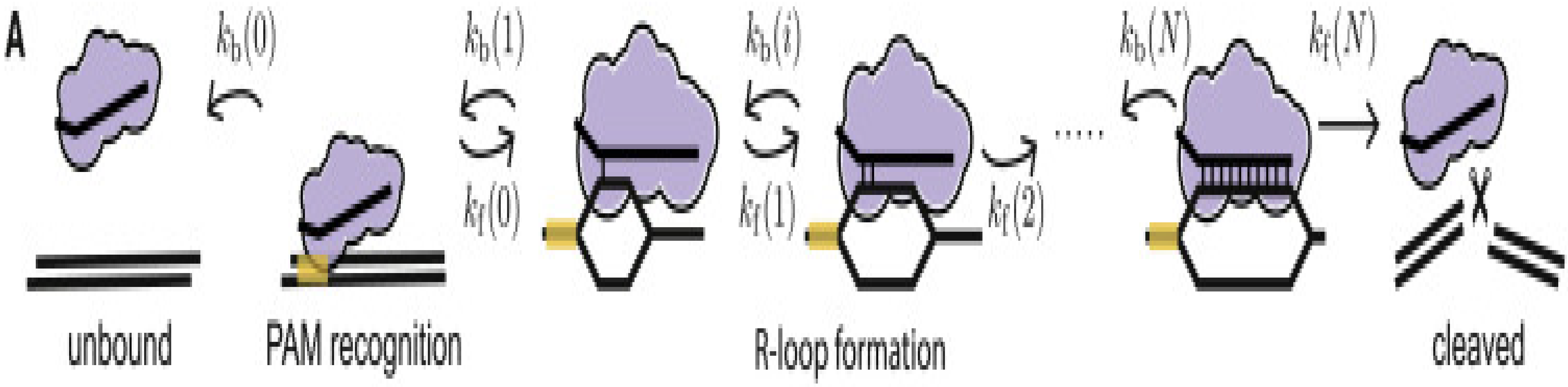
Cas9 Variants

SpCas9 variant	Mutations (relative to SpCas9)	PAM sequence
D1135E variant	D1135E	NGG
VQR variant	D1135V, R1335Q and T1337R	NGAN or NGNG
EQR variant	D1135E, R1335Q and T1337R	NGAG
VRER variant	D1135V, G1218R, R1335E and T1337R	NGCG

Cas9 Orthologs

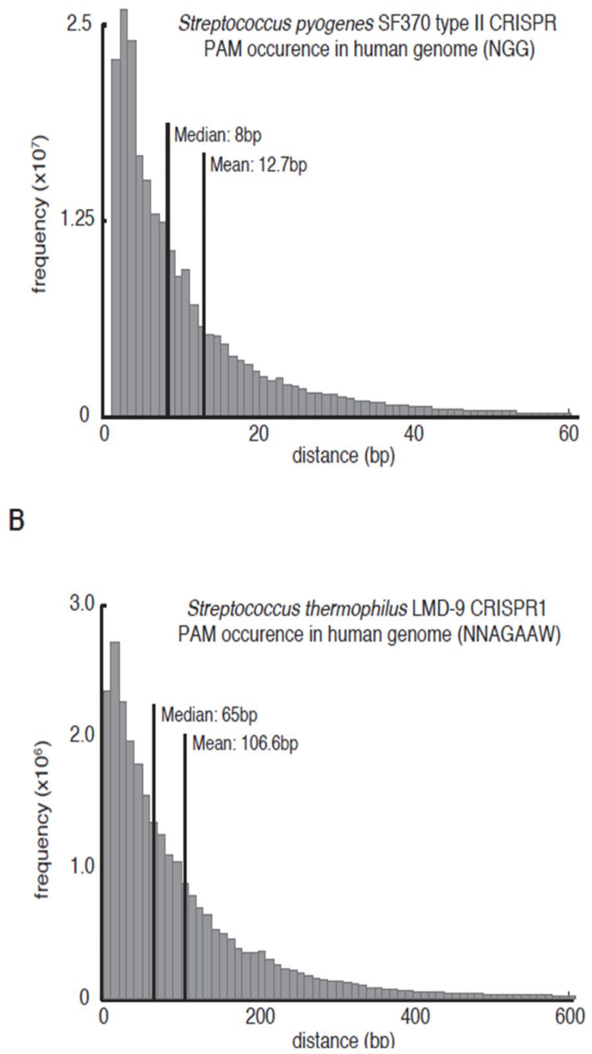
Cas9 species	PAM sequence ’)
Streptococcus pyogenes (Sp)	3' NGG
Staphylococcus aureus (Sa)	NGRRT or NGRRN
Neisseria meningitidis (Nm or Nme)	NNNNGATT
Campylobacter jejuni (Cj)	NNNNRYAC
Streptococcus thermophilus (St)	NNAGAAW
Treponema denticola (Td)	NAAAAC
~20 additional Cas9 species	PAM sequence may not be characterized

Key activities of Cas9



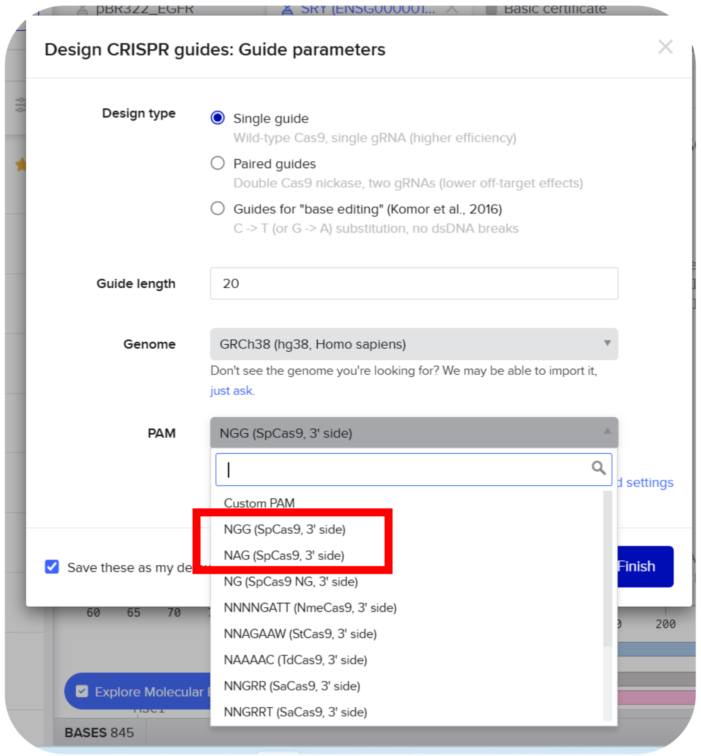
https://www.sciencedirect.com/science/article/pii/S2211124718300779

PAM occurrence on the human genome



	NGG		NNAGAAW	
Chr	median	mean	median	mean
1	7	12.8	67	115.8
2	8	12.7	64	100.8
3	8	13.0	63	98.5
4	9	14.0	61	94.5
5	8	13.1	63	97.9
6	8	13.1	63	98.5
7	8	12.4	64	102.9
8	8	12.8	64	100.9
9	7	13.9	65	120.5
10	7	12.1	66	107.0
11	7	12.0	65	105.8
12	8	12.4	65	103.5
13	8	13.6	62	94.6
14	8	12.0	65	101.5
15	7	11.5	68	107.7
16	7	11.7	74	136.8
17	6	10.3	76	127.9
18	8	13.4	63	101.8
19	6	9.4	82	145.4
20	7	11.1	72	121.8
21	7	13.4	64	111.4
22	6	9.2	85	140.3
X	8	13.2	63	99.0
Y	8	29.2	62	223.7

Benchling



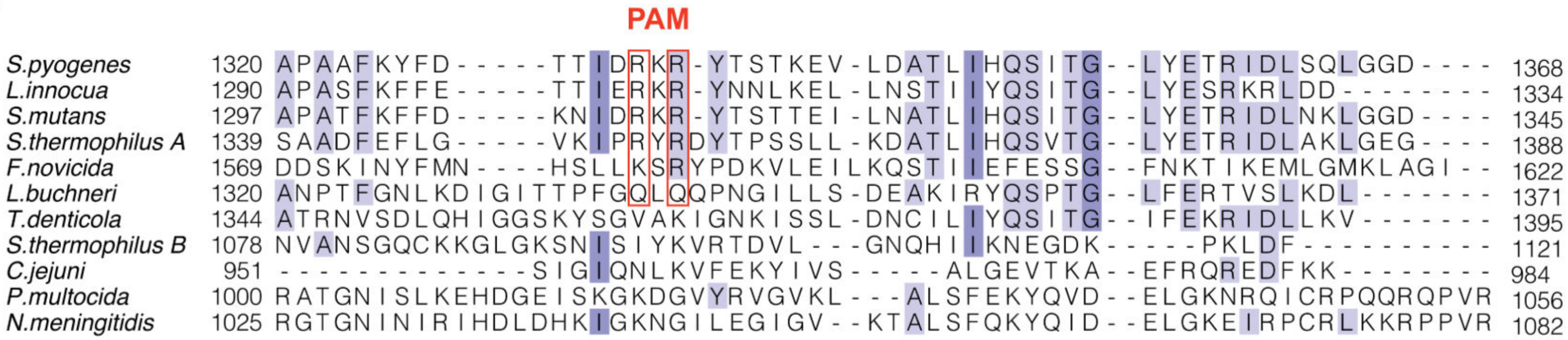
https://pmc.ncbi.nlm.nih.gov/articles/PMC4540238/

Cas9 proteins and PAMS

a

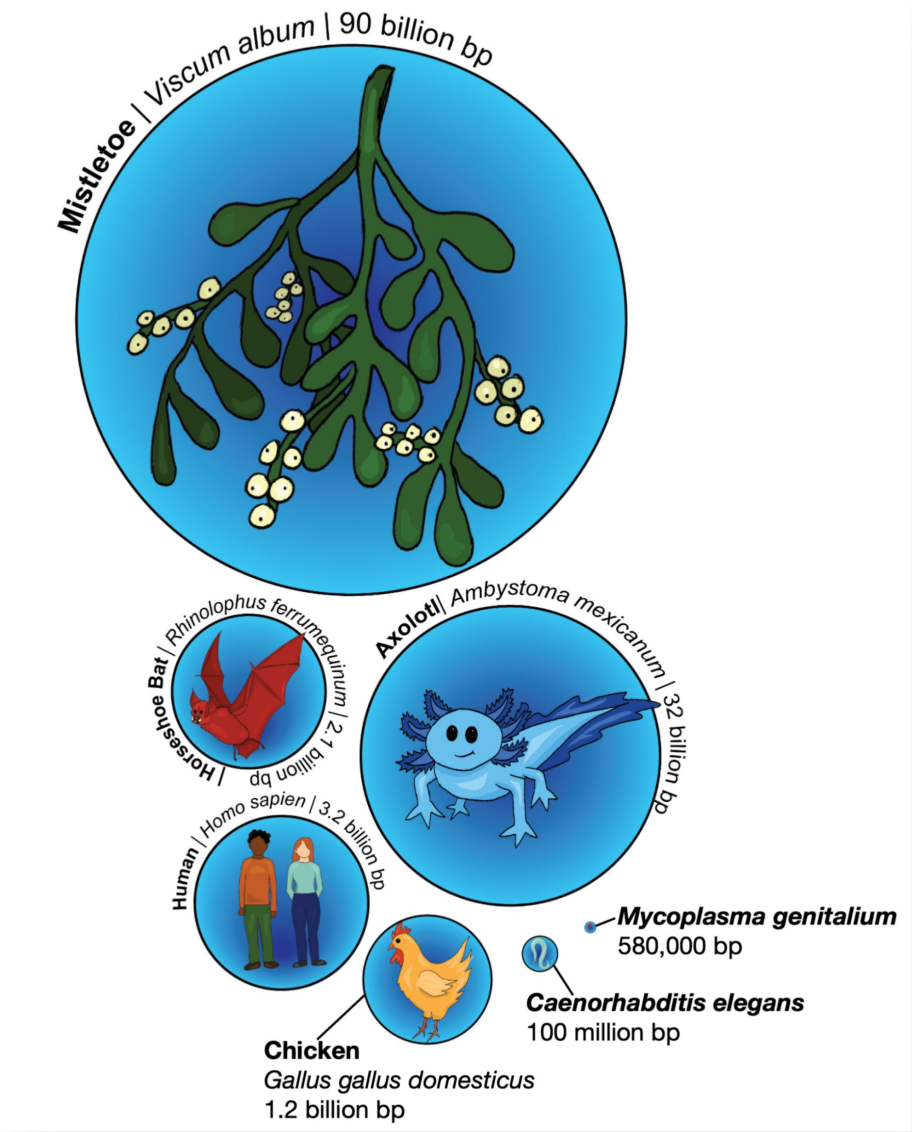
Cas9	PAM	Type
Streptococcus pyogenes	NGG	II-A
Streptococcus mutans	NGG	II-A
Listeria innocua	NGG	II-A
Streptococcus thermophilus A (CRISPR3)	NGGNG	II-A
Francisella novicida	NG	II-B
Lactobacillus buchneri	NAAAAN	II-A
Treponema denticola	NAAAAN	II-A
Streptococcus thermophilus B (CRISPR1)	NNAAAAW	II-A
Campylobacter jejuni	NNNNACA	II-C
Pasteurella multocida	GNNNCNNA	II-C
Neisseria meningitidis	NNNNGATT	II-C

b



Nature. 2014 September 25; 513(7519)

Genome Diversity



Wellcome Sanger Blog

Mistletoe (*Viscum album*): ~ 90 billion base pairs of DNA (bp), 10 chromosome pairs, 39,000 genes

Axolotl (*Ambystoma mexicanum*): ~ 32 billion base pairs, 14 chromosome pairs

Human (*Homo sapiens*): ~ 3.2 billion base pairs, 23 chromosome pairs, 21,000 genes

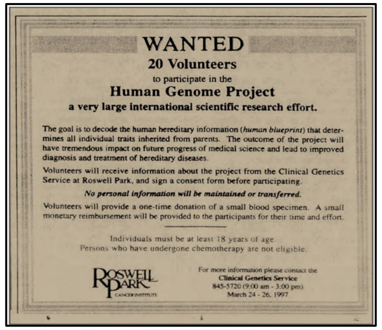
Horseshoe bat (*Rhinolophus ferrumequinum*): ~ 2.1 billion base pairs, 19,000 genes.

Chicken (*Gallus gallus*): ~ 1.2 billion base-pairs, 39 chromosomes (most of which are 'microchromosomes') 20,000 genes

The *Caenorhabditis elegans* worm: ~ 100 million base pairs, 6 chromosome pairs, 20,500 genes

Bacteria *Mycoplasma genitalium*: ~ 580,000 base pairs, 1 chromosome, 470 genes

The human genome 1997- 2003


WANTED

20 Volunteers

to participate in the

Human Genome Project

a very large international scientific research effort.

The goal is to decode the human hereditary information (*human blueprint*) that determines all individual traits inherited from parents. The outcome of the project will have tremendous impact on future progress of medical science and lead to improved diagnosis and treatment of hereditary diseases.

Volunteers will receive information about the project from the Clinical Genetics Service at Roswell Park, and sign a consent form before participating.

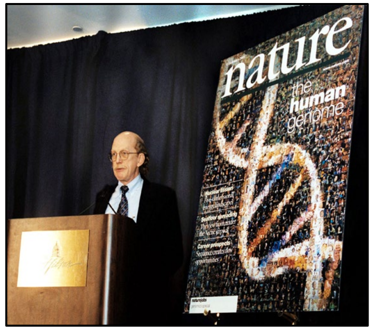
No personal information will be maintained or transferred.

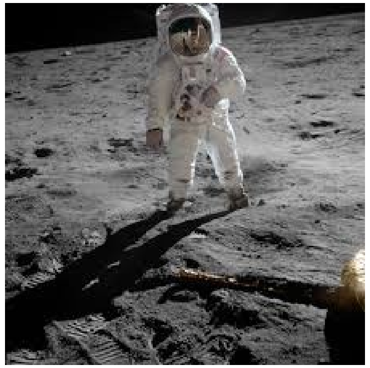
Volunteers will provide a one-time donation of a small blood specimen. A small monetary reimbursement will be provided to the participants for their time and effort.

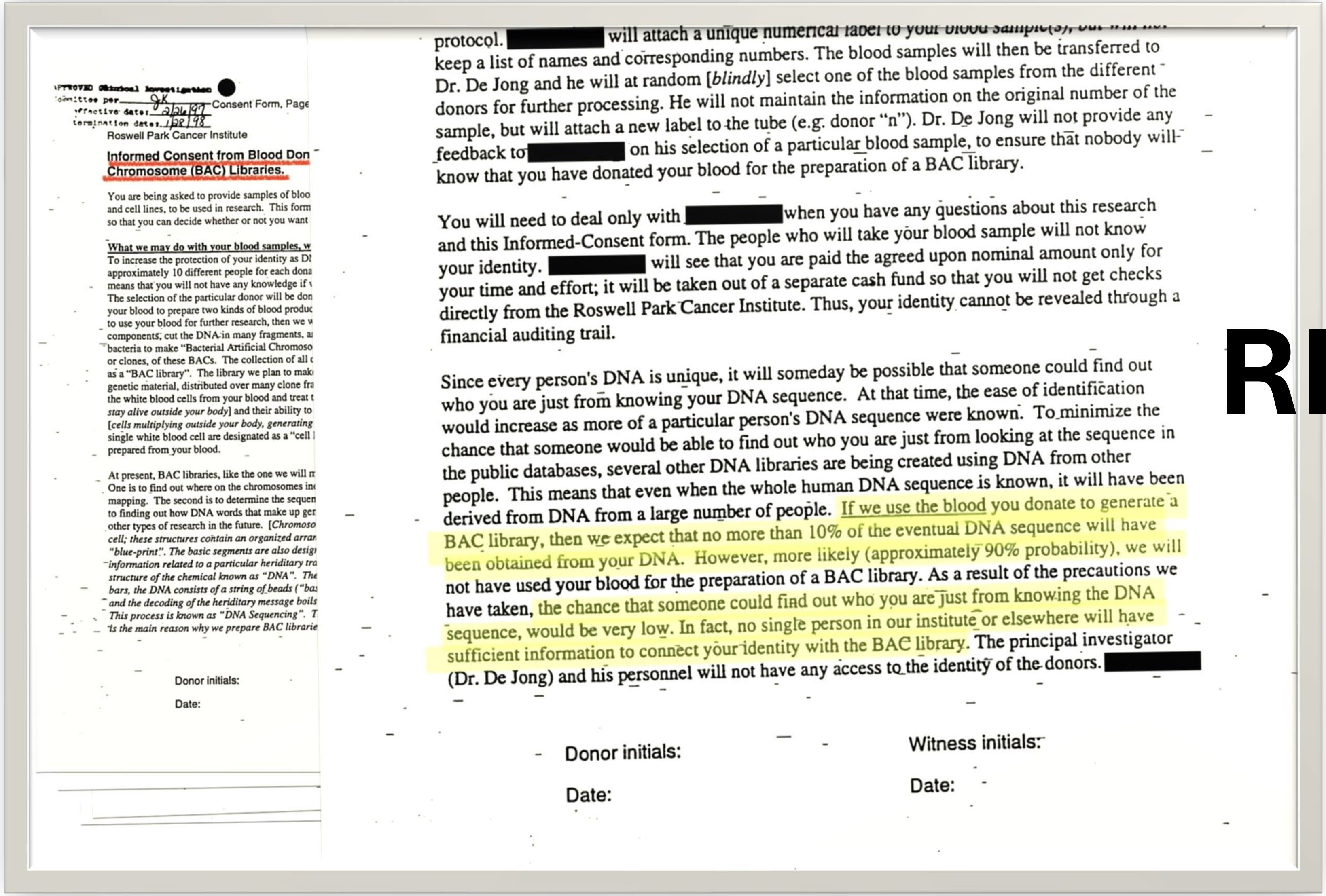
Individuals must be at least 18 years of age.
Persons who have undergone chemotherapy are not eligible.

ROSWELL PARK CANCER INSTITUTE

For more information please contact the
Clinical Genetics Service
845-5720 (9:00 am - 3:00 pm)
March 24 - 26, 1997






APPROVED Clinical Investigation
Committee per JK
effective date: 2/26/97
termination date: 1/28/98

Consent Form, Page

Roswell Park Cancer Institute

Informed Consent from Blood Don
Chromosome (BAC) Libraries.

You are being asked to provide samples of bloo
and cell lines, to be used in research. This form
so that you can decide whether or not you want

What we may do with your blood samples, w
To increase the protection of your identity as DI
approximately 10 different people for each dona
means that you will not have any knowledge if
The selection of the particular donor will be don
your blood to prepare two kinds of blood produc
to use your blood for further research, then we w
components, cut the DNA in many fragments, a
bacteria to make "Bacterial Artificial Chromoso
or clones, of these BACs. The collection of all
as a "BAC library". The library we plan to mak
genetic material, distributed over many clone fra
the white blood cells from your blood and treat t
stay alive outside your body] and their ability to
[*cells multiplying outside your body, generating*
single white blood cell are designated as a "cell
prepared from your blood.

At present, BAC libraries, like the one we will m
One is to find out where on the chromosomes in
mapping. The second is to determine the sequen
to finding out how DNA words that make up ger
other types of research in the future. [*Chromoso*
cell; these structures contain an organized arran
"blue-print". The basic segments are also desig
information related to a particular heriditary tra
structure of the chemical known as "DNA". The
bars, the DNA consists of a string of beads ("ba
and the decoding of the heriditary message boils
This process is known as "DNA Sequencing". T
is the main reason why we prepare BAC librarie

Donor initials:

Date:

protocol. [redacted] will attach a unique numerical label to your blood sample(s), but will not keep a list of names and corresponding numbers. The blood samples will then be transferred to Dr. De Jong and he will at random [*blindly*] select one of the blood samples from the different donors for further processing. He will not maintain the information on the original number of the sample, but will attach a new label to the tube (e.g. donor "n"). Dr. De Jong will not provide any feedback to [redacted] on his selection of a particular blood sample, to ensure that nobody will know that you have donated your blood for the preparation of a BAC library.

You will need to deal only with [redacted] when you have any questions about this research and this Informed-Consent form. The people who will take your blood sample will not know your identity. [redacted] will see that you are paid the agreed upon nominal amount only for your time and effort; it will be taken out of a separate cash fund so that you will not get checks directly from the Roswell Park Cancer Institute. Thus, your identity cannot be revealed through a financial auditing trail.

Since every person's DNA is unique, it will someday be possible that someone could find out who you are just from knowing your DNA sequence. At that time, the ease of identification would increase as more of a particular person's DNA sequence were known. To minimize the chance that someone would be able to find out who you are just from looking at the sequence in the public databases, several other DNA libraries are being created using DNA from other people. This means that even when the whole human DNA sequence is known, it will have been derived from DNA from a large number of people. If we use the blood you donate to generate a BAC library, then we expect that no more than 10% of the eventual DNA sequence will have been obtained from your DNA. However, more likely (approximately 90% probability), we will not have used your blood for the preparation of a BAC library. As a result of the precautions we have taken, the chance that someone could find out who you are just from knowing the DNA sequence, would be very low. In fact, no single person in our institute or elsewhere will have sufficient information to connect your identity with the BAC library. The principal investigator (Dr. De Jong) and his personnel will not have any access to the identity of the donors. [redacted]

Donor initials:

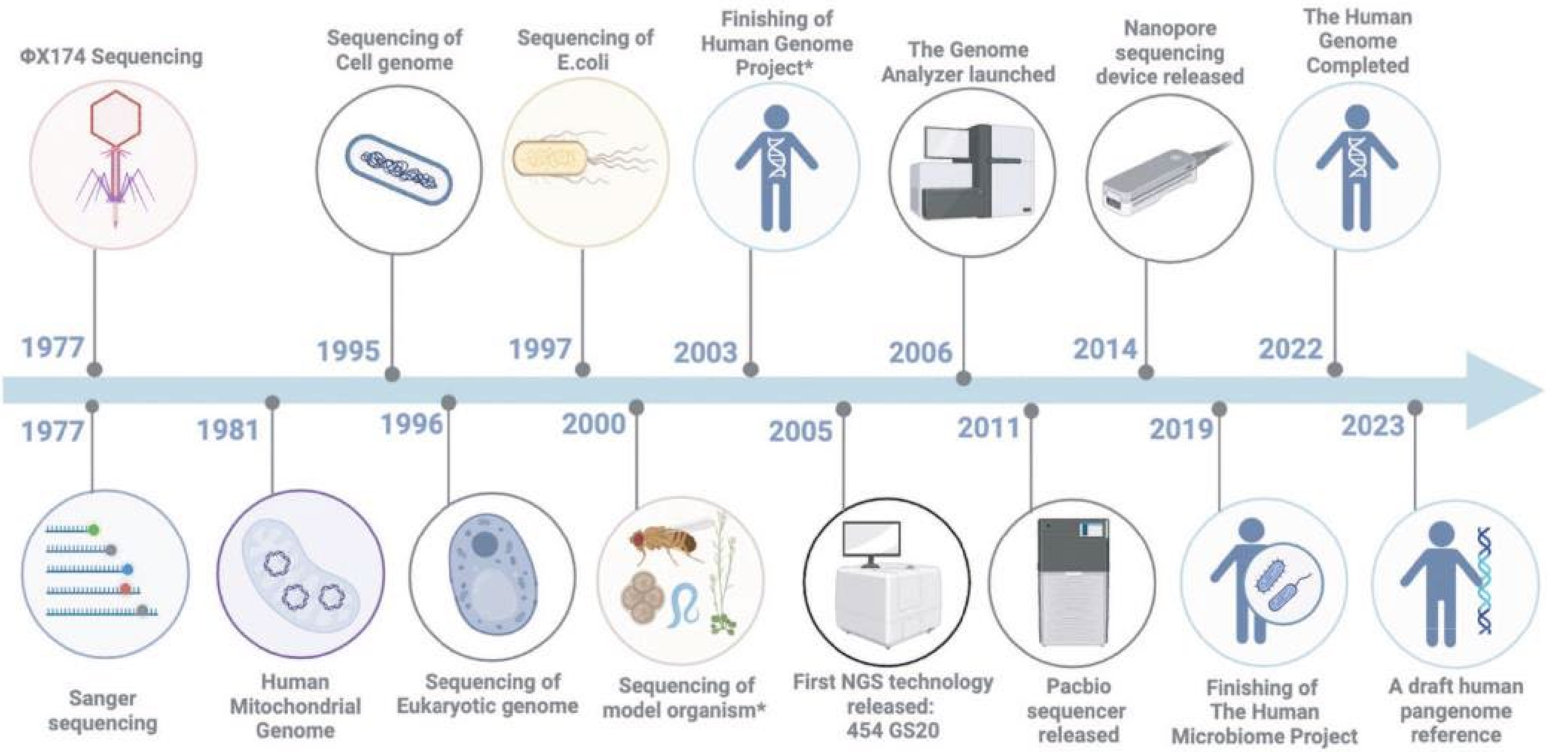
Date:

Witness initials:

Date:

RP11

Human Pangenome Reference Consortium



Freeze date	GENCODE release	Reference release?	Release date	Genome assembly version	UCSC version	Notes
03.2020	35	Y	08.2020	GRCh38	-	re-merge with new Havana annotation, updated Ensembl gene set
11.2019	34	N	04.2020	GRCh38	34	re-merge with new Havana annotation, updated Ensembl gene set
08.2019	33	N	01.2020	GRCh38	33	re-merge with new Havana annotation, updated Ensembl gene set
05.2019	32	N	09.2019	GRCh38	32 (current for GRCh38)	re-merge with new Havana annotation, updated Ensembl gene set
02.2019	31	N	06.2019	GRCh38	31	re-merge with new Havana annotation, updated Ensembl gene set
11.2018	30	N	04.2019	GRCh38	30	re-merge with new Havana annotation, updated Ensembl gene set
05.2018	29	N	10.2018	GRCh38	29	re-merge with new Havana annotation, updated Ensembl gene set

https://www.sciencedirect.com/science/article/pii/S0888754324000636?via%3Dihub

Human Pangenome Reference Consortium

Funded by: **NIH NHGRI (National Human Genome Research Institute)**

University of California, Santa Cruz (UCSC)
The Rockefeller University
The Broad Institute
Washington University in St. Louis
Baylor College of Medicine

Cancer Drug Discovery

PCAWG
PanCancer Analysis of WHOLE GENOMES

wellcome sanger institute

cell model passports

10 YEAR ANNIVERSARY
Open Targets
2014-2024

Project Achilles

CCLE Cancer Cell Line Encyclopedia

depmap

BROAD INSTITUTE

project score

cancer dependency map

Seminal articles in CRISPR/Cas9

hhmi | Howard Hughes Medical Institute

Published as: *Science*. 2012 August 17; 337(6096): 816–821.

A programmable dual RNA-guided DNA endonuclease in adaptive bacterial immunity

Martin Jinek[#1,2], **Krzysztof Chylinski**[#3,4], **Ines Fonfara**[4], **Michael Hauer**[2,5], **Jennifer A. Doudna**[1,2,6,7,*], and **Emmanuelle Charpentier**[4,*]

[1]Howard Hughes Medical Institute, University of California, Berkeley, California 94720, USA. [2]Department of Molecular and Cell Biology, University of California, Berkeley, California 94720, USA. [3]Max F. Perutz Laboratories, University of Vienna, A-1030 Vienna, Austria. [4]The Laboratory for Molecular Infection Medicine Sweden (MIMS), Umeå Centre for Microbial Research (UCMR), Department of Molecular Biology, Umeå University, S-90187 Umeå, Sweden. [5]Present address: Friedrich Miescher Institute for Biomedical Research, 4058 Basel, Switzerland. [6]Department of Chemistry, University of California, Berkeley, California 94720, USA. [7]Physical Biosciences Division, Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA.

[#] These authors contributed equally to this work.

Abstract

CRISPR/Cas systems provide bacteria and archaea with adaptive immunity against viruses and plasmids by using crRNAs to guide the silencing of invading nucleic acids. We show here that in a subset of these systems, the mature crRNA base-paired to *trans*-activating tracrRNA forms a two-RNA structure that directs the CRISPR-associated protein Cas9 to introduce double-stranded (ds) breaks in target DNA. At sites complementary to the crRNA-guide sequence, the Cas9 HNH nuclease domain cleaves the complementary strand while the Cas9 RuvC-like domain cleaves the non-complementary strand. The dual-tracrRNA:crRNA, when engineered as a single RNA chimera, also directs sequence-specific Cas9 dsDNA cleavage. Our study reveals a family of

PROTOCOL

Genome engineering using the CRISPR-Cas9 system

F Ann Ran[1–5,8], Patrick D Hsu[1–5,8], Jason Wright[1], Vineeta Agarwala[1,6,7], David A Scott[1–4] & Feng Zhang[1–4]

[1]Broad Institute of Massachusetts Institute of Technology (MIT) and Harvard, Cambridge, Massachusetts, USA. [2]McGovern Institute for Brain Research, Cambridge, Massachusetts, USA. [3]Department of Brain and Cognitive Sciences, MIT, Cambridge, Massachusetts, USA. [4]Department of Biological Engineering, MIT, Cambridge, Massachusetts, USA. [5]Department of Molecular and Cellular Biology, Harvard University, Cambridge, Massachusetts, USA. [6]Program in Biophysics, Harvard University, MIT, Cambridge, Massachusetts, USA. [7]Harvard-MIT Division of Health Sciences and Technology, MIT, Cambridge, Massachusetts, USA. [8]These authors contributed equally to this work. Correspondence should be addressed to F.Z. (zhang@broadinstitute.org).

Published online 24 October 2013; doi:10.1038/nprot.2013.143

Targeted nucleases are powerful tools for mediating genome alteration with high precision. The RNA-guided Cas9 nuclease from the microbial clustered regularly interspaced short palindromic repeats (CRISPR) adaptive immune system can be used to facilitate efficient genome engineering in eukaryotic cells by simply specifying a 20-nt targeting sequence within its guide RNA. Here we describe a set of tools for Cas9-mediated genome editing via nonhomologous end joining (NHEJ) or homology-directed repair (HDR) in mammalian cells, as well as generation of modified cell lines for downstream functional studies. To minimize off-target cleavage, we further describe a double-nicking strategy using the Cas9 nickase mutant with paired guide RNAs. This protocol provides experimentally derived guidelines for the selection of target sites, evaluation of cleavage efficiency and analysis of off-target activity. Beginning with target design, gene modifications can be achieved within as little as 1–2 weeks, and modified clonal cell lines can be derived within 2–3 weeks.

INTRODUCTION

The ability to engineer biological systems and organisms holds enormous potential for applications across basic science, medicine and biotechnology. Programmable sequence-specific endonucleases that facilitate precise editing of endogenous genomic loci are now enabling systematic interrogation of genetic elements and causal genetic variations[1,2] in a broad range of species, including those that have not previously been genetically tractable[3–6]. A number of genome editing technologies have emerged in recent years, including zinc-finger nucleases (ZFNs)[7–10], transcription activator–like effector nucleases (TALENs)[10–17] and the RNA-guided CRISPR-Cas nuclease system[18–25]. The first two technologies use a strategy of tethering endonuclease catalytic domains to modular DNA-binding proteins for inducing targeted DNA double-stranded breaks (DSBs) at specific genomic loci. By contrast, Cas9 is a nuclease guided by small RNAs through Watson-Crick base pairing with target DNA[26–28] (**Fig. 1**), representing a system that is markedly easier to design, highly specific, efficient and well-suited for high-throughput and multiplexed gene editing for a variety of cell types and organisms.

Precise genome editing using engineered nucleases

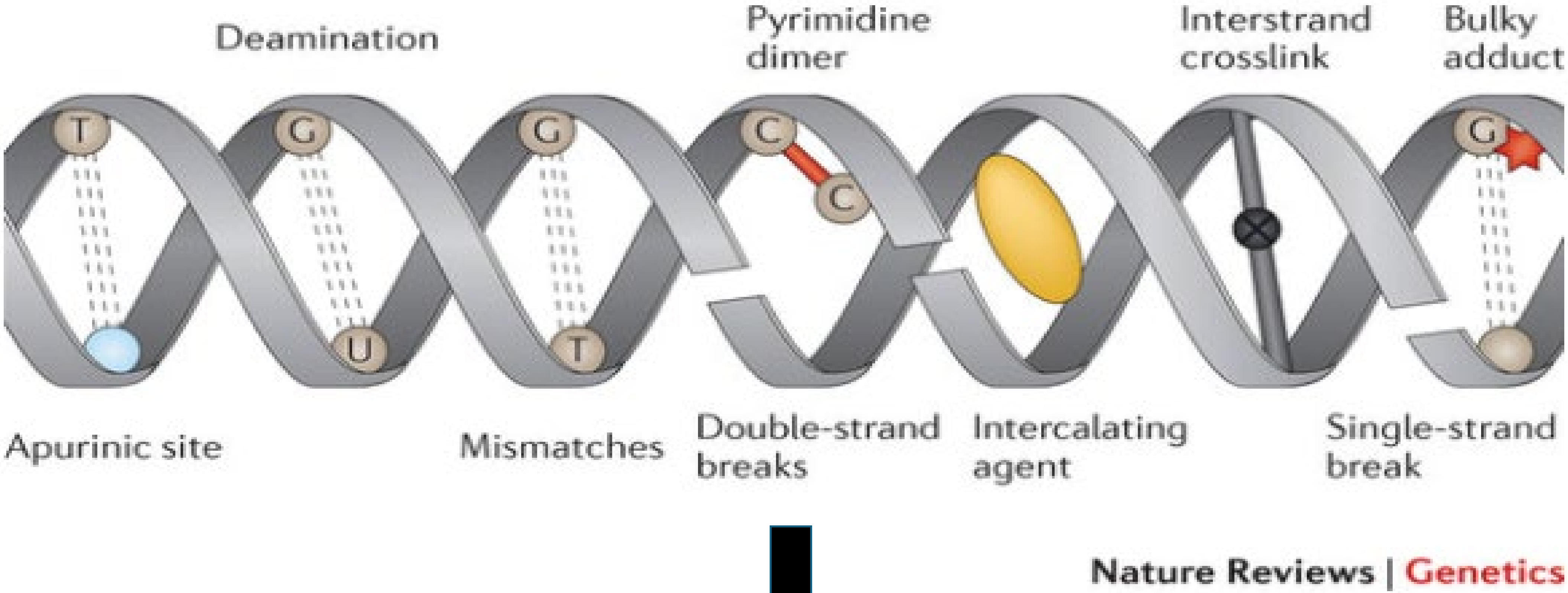
Similarly to ZFNs and TALENs, Cas9 promotes genome editing by stimulating a DSB at a target genomic locus[29,30]. Upon cleavage by Cas9, the target locus typically undergoes one of two major pathways for DNA damage repair (**Fig. 2**): the error-prone NHEJ

frequencies than NHEJ, it can be leveraged to generate precise, defined modifications at a target locus in the presence of an exogenously introduced repair template. The repair template can either be in the form of conventional double-stranded DNA targeting constructs with homology arms flanking the insertion sequence, or single-stranded DNA oligonucleotides (ssODNs). The latter provides an effective and simple method for making small edits in the genome, such as the introduction of single-nucleotide mutations for probing causal genetic variations[32]. Unlike NHEJ, HDR is generally active only in dividing cells, and its efficiency can vary widely depending on the cell type and state, as well as the genomic locus and repair template[33].

Cas9: an RNA-guided nuclease for genome editing

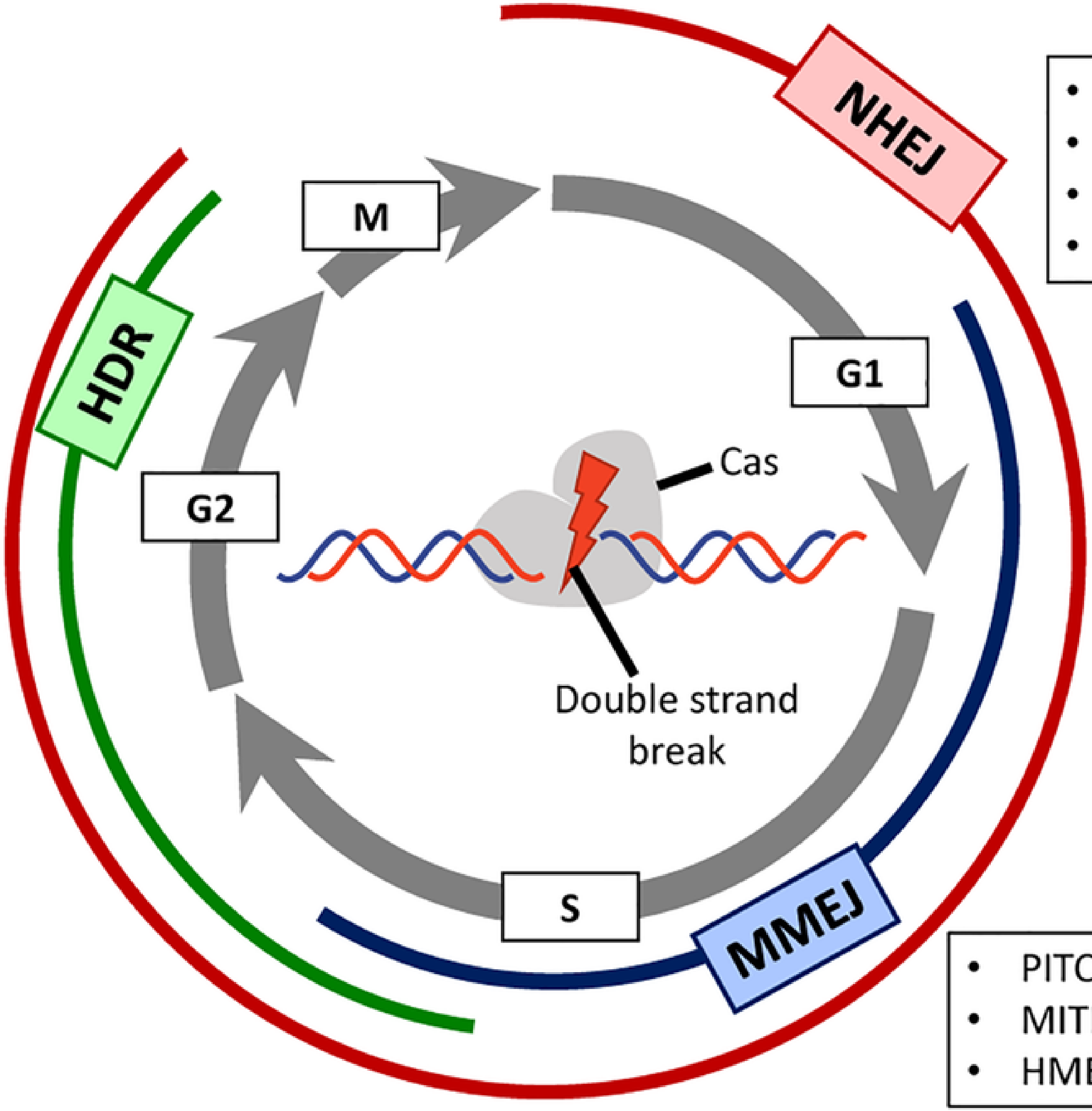
CRISPR-Cas is a microbial adaptive immune system that uses RNA-guided nucleases to cleave foreign genetic elements[18–21,26]. Three types (I–III) of CRISPR systems have been identified across a wide range of bacterial and archaeal hosts, wherein each system comprises a cluster of CRISPR-associated (*Cas*) genes, noncoding RNAs and a distinctive array of repetitive elements (direct repeats). These repeats are interspaced by short variable sequences[20] derived from exogenous DNA targets known as protospacers, and together they constitute the CRISPR RNA (crRNA) array. Within the DNA target, each protospacer is always associated with a protospacer adjacent motif (PAM), which can vary depending on the specific CRISPR system[34–36].

DNA damage



Free radical species are generated either as a by-product of metabolism or through exposure to exogenous physical agents, such as ionizing radiation, which can induce the formation of double-strand breaks.

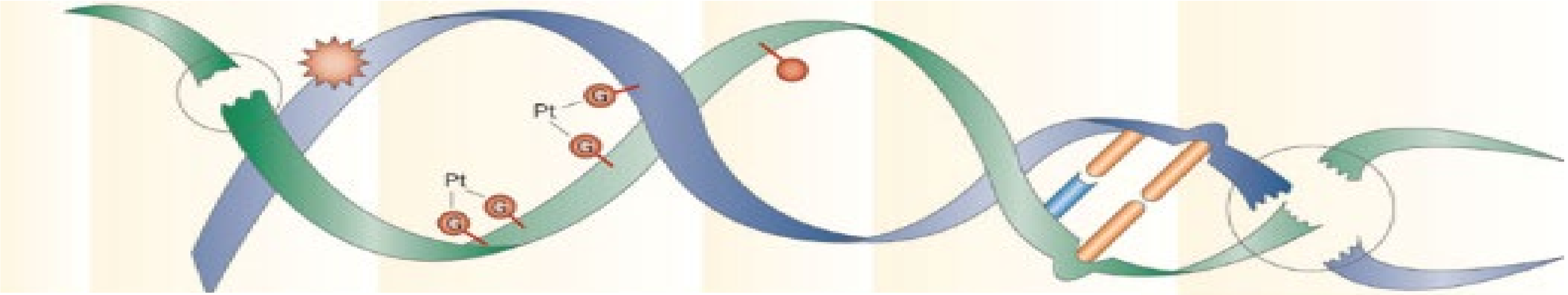
DNA repair pathways and Gene Editing



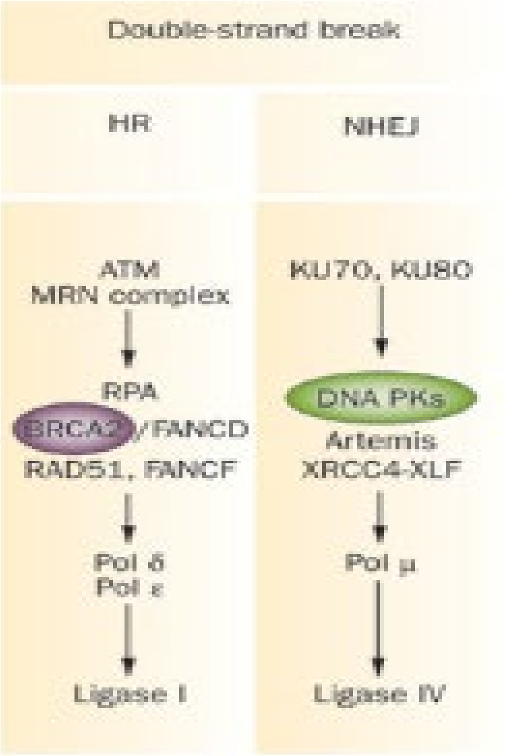
- HITI
- MITI
- CRISPaint
- CRISPR-HOT

- PITCh
- MITI
- HMEJ

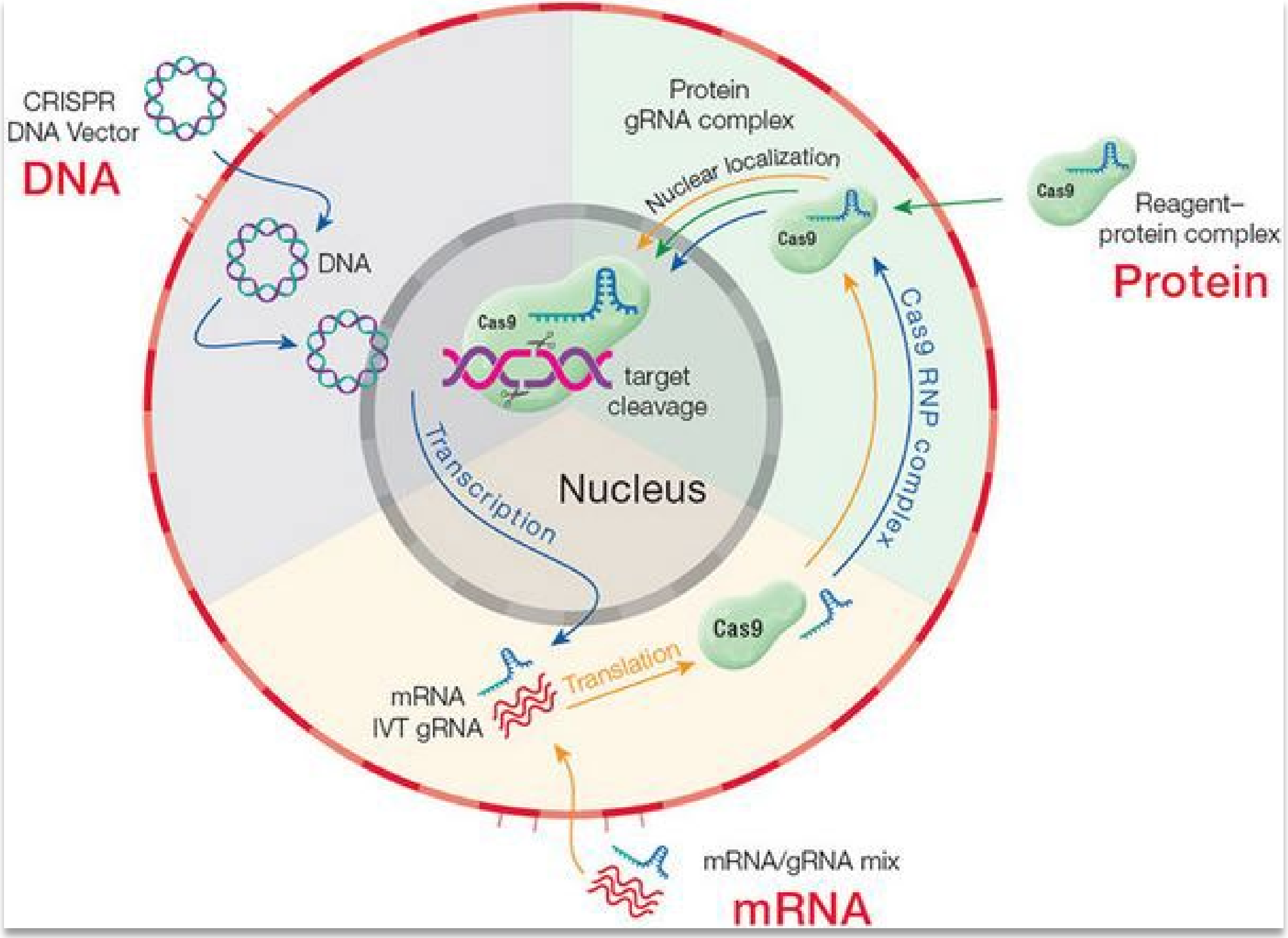
Double Strand Break damage



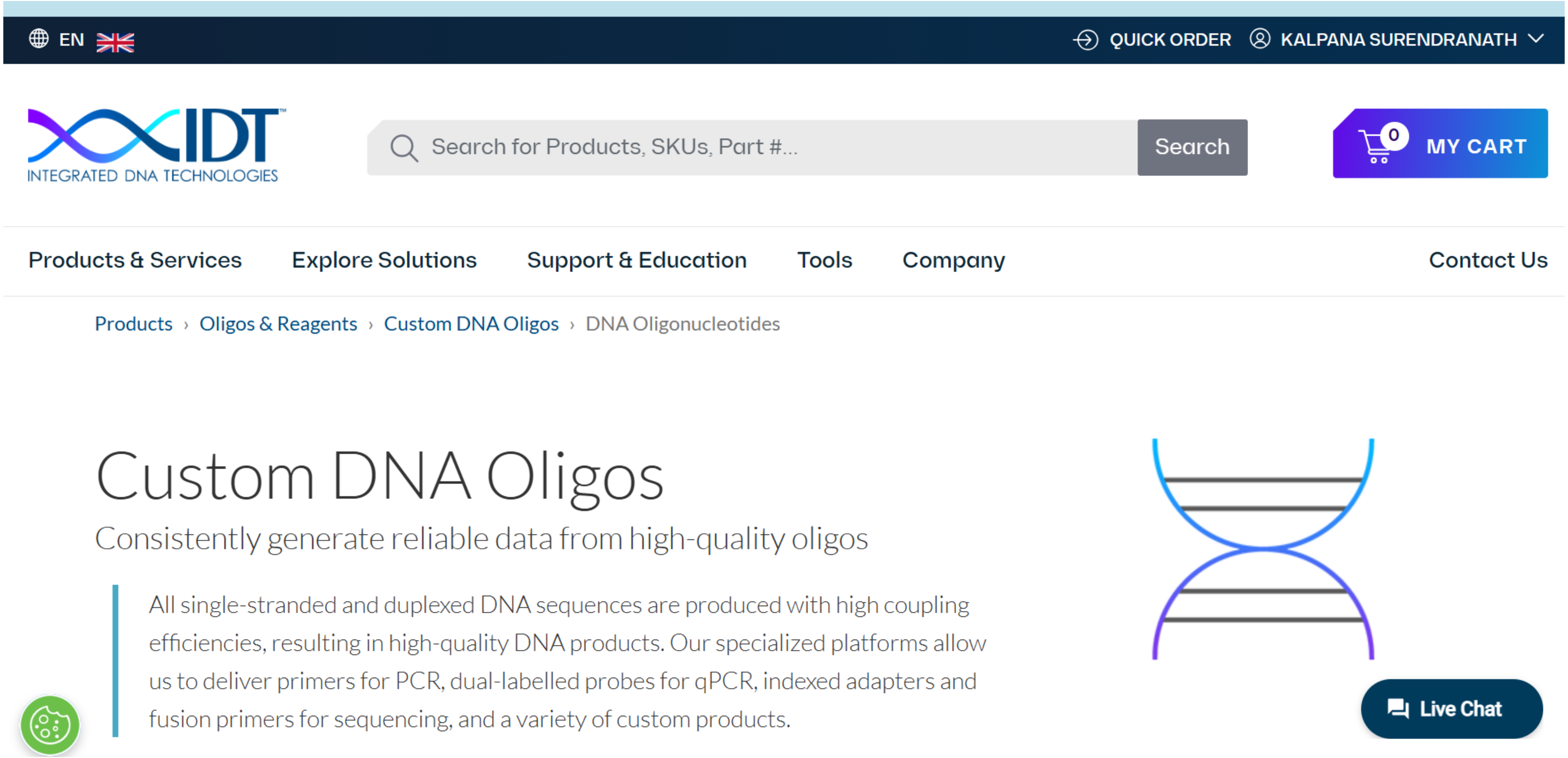
Homologous recombination Vs Non-Homologous end joining



Roadmap of CRISPR tools



Vendor options/tools



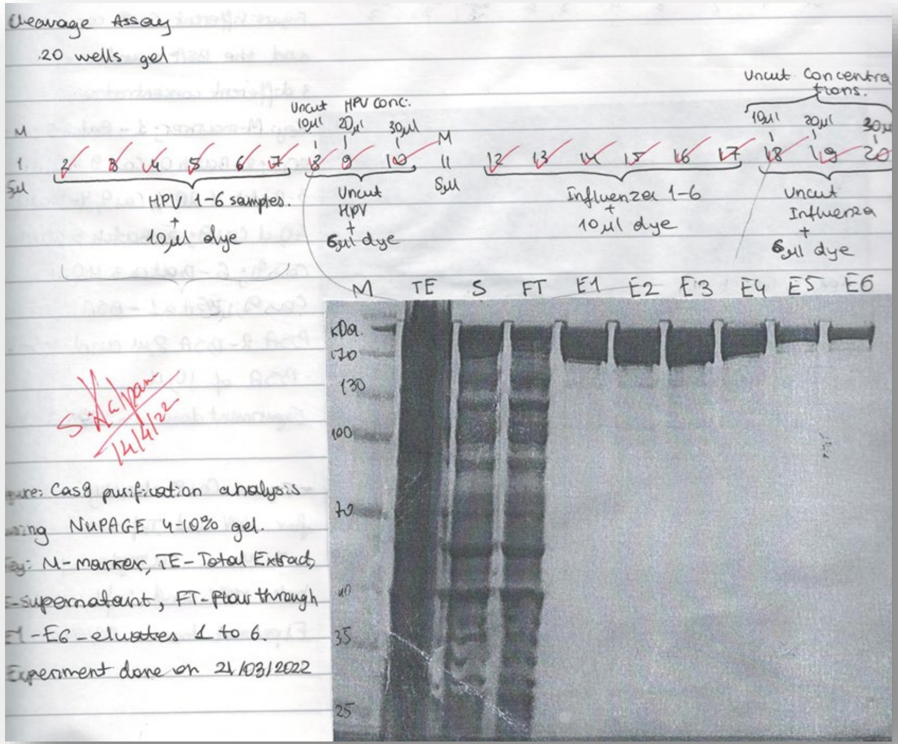
Vendor options/tools

Alt-R gRNA OPTIONS FOR *S.p.* Cas9 & *A.s.* Cas12a

Guide RNAs	Cas9 guide RNAs			Cas12a guide RNAs
Structure	Alt-R 2-part	Alt-R 2-part XT	Alt-R sgRNA	Alt-R Cas12a crRNA
gRNA format	Alt-R CRISPR-Cas9 crRNA & tracrRNA	Alt-R CRISPR-Cas9 crRNA XT & tracrRNA	Alt-R CRISPR-Cas9 sgRNA	Alt-R CRISPR-Cas12a crRNA
Components	crRNA tracrRNA	crRNA XT tracrRNA	sgRNA	crRNA
Size (nt)	36 67	36 67	100	40–44 (41 nt recommended)
Annealing required	Yes	Yes	No	No
Stability	++	+++	++++	+++
Applications	• Cas9-expressing cells • RNP in most cell types	• Co-delivery with Cas9 plasmid/Cas9 mRNA • RNP under diffcult expermental condiitions (e.g., high nuclease environments)		• KO/KI, RNP in most cell types • Cas12a-expressing cells

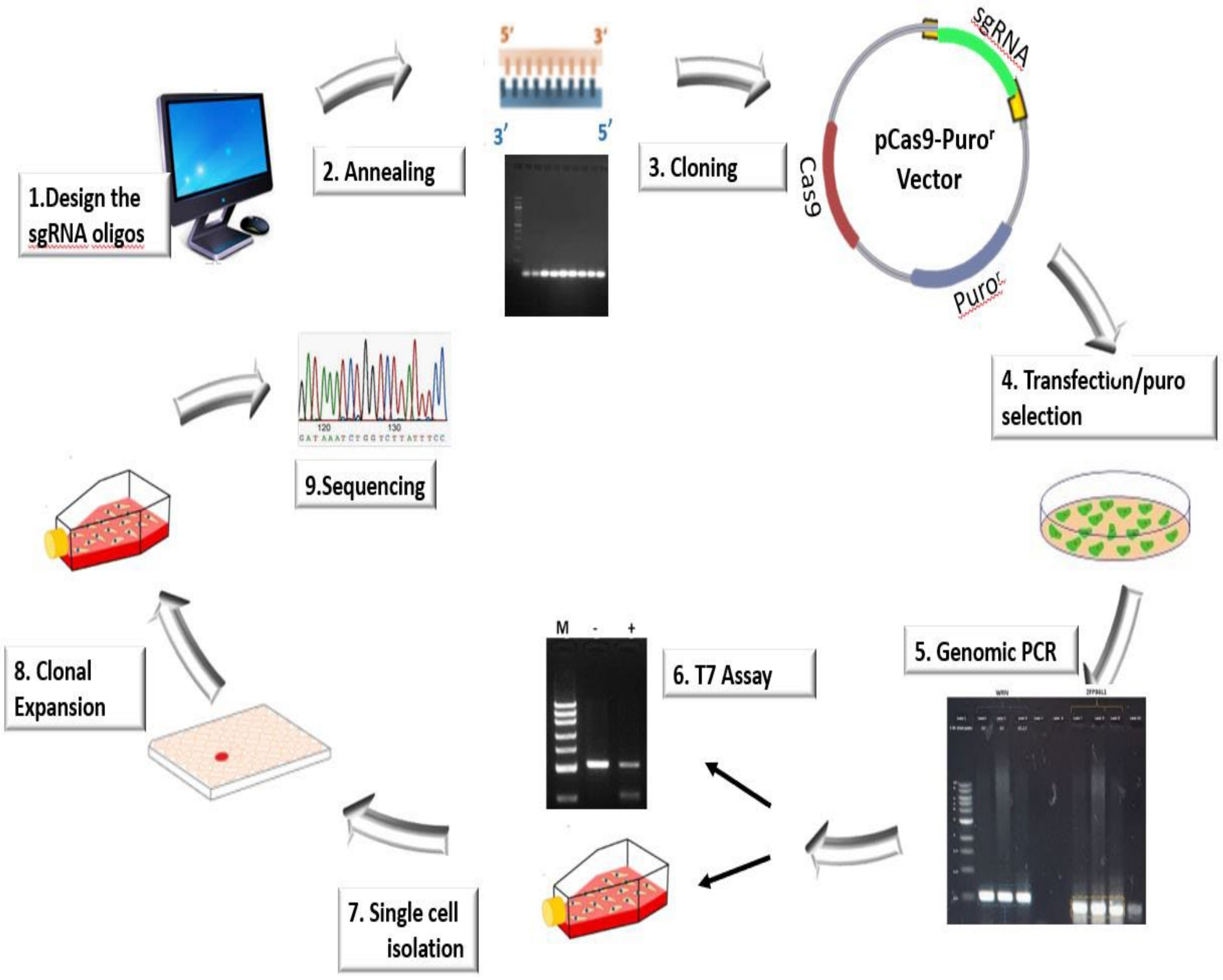
Quantity	Product	Catalog #	Price
0	Alt-R™ S.p. Cas9 Nuclease V3, 500 µg	1081059	£653.00 GBP

https://eu.idtdna.com/site/order/stock/partnbr/1081059



Two-step enrichment of Cas9

Workflow for Knock out generation



Gene Editing Workflow- Stage 1

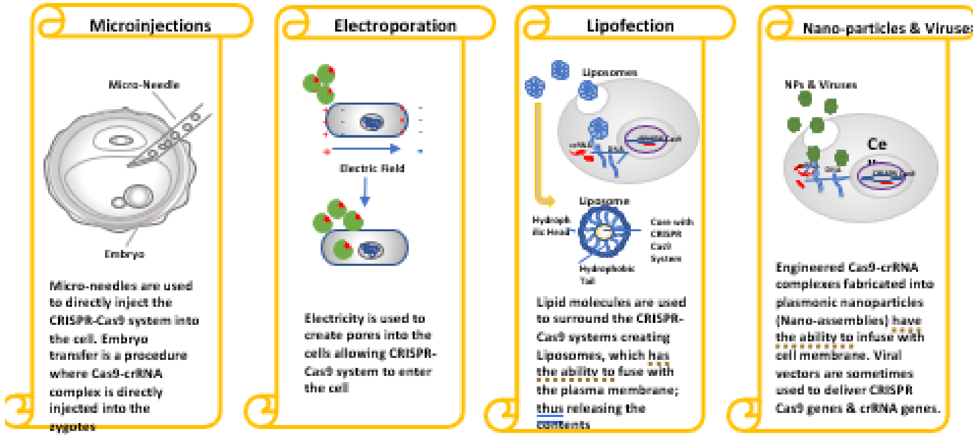
1. Design and selection of guide sequences (online algorithm)
2. Synthesis of single stranded guide oligos
3. Clone into CRISPR/Cas9 expression vector
4. Sanger Sequencing
5. Plasmids purification

Stage1

sgRNA synthesis and RNP complex formation

6. Transfect cells
7. Selection e.g. Puromycin
8. Clonal Isolation and expansion

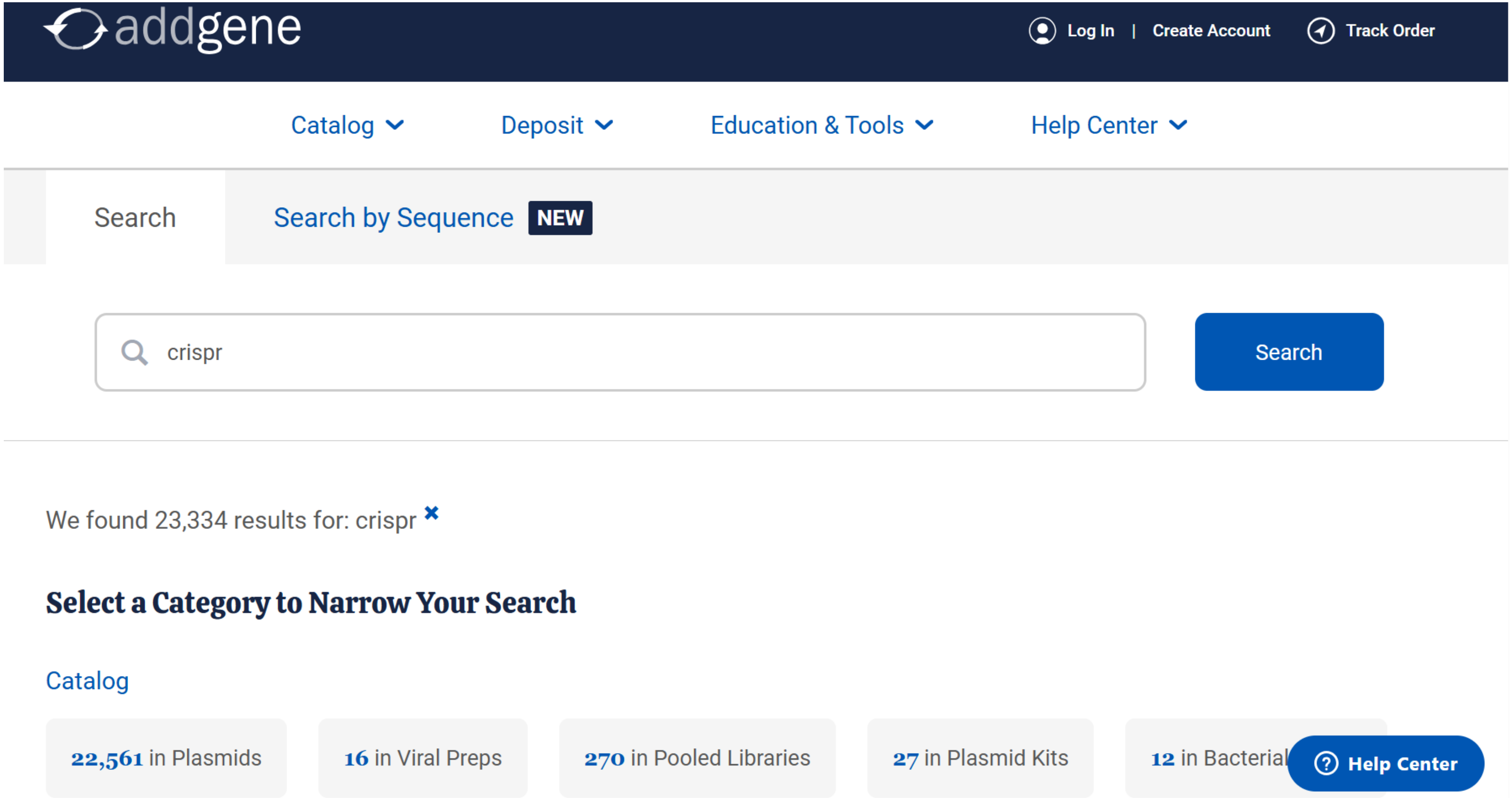
Stage2
~ 2 months



9. Functional characterization and phenotype and genotype analysis

Stage 3

Plasmids tools



www.addgene.org

Early days…

http://crispr.mit.edu

Haeussler, M., Schönig, K., Eckert, H. *et al*. Evaluation of off-target and on-target scoring algorithms and integration into the guide RNA selection tool CRISPOR. *Genome Biol* **17,** 148 (2016).

Guide Design Webtools

Algorithm	Editing	URL
Cas-Designer	CRISPR-Cas mediated editing	http://www.rgenome.net/cas-designer/
Cas-OFFinder		http://www.rgenome.net/cas-offinder/
CHOPCHOP		https://chopchop.cbu.uib.no/
CRISPR-P 2.0		http://crispr.hzau.edu.cn/CRISPR2/
CRISPOR		http://crispor.tefor.net/
CRISPRdirect		https://crispr.dbcls.jp/
E-CRISP		http://www.e-crisp.org/
CCTop		http://crispr.cos.uni-heidelberg.de/
DeepCpf1		http://deepcrispr.info/
DeepSpCas9		http://deepcrispr.info/DeepSpCas9
DeepHF		http://www.DeepHF.com/
CINDEL		http://big.hanyang.ac.kr/cindel
SSC		http://cistrome.org/SSC/
Prediction web tool		https://crispr.ml
CRISPRscan		https://www.crisprscan.org
Microhomology-Predictor		http://www.rgenome.net/michcalculator
FORECasT		https://partslab.sanger.ac.uk/FORECasT
inDelphi		https://indelphi.giffordlab.mit.edu
TIDE		http://shinyapps.datacurators.nl
Cas-Analyzer		http://www.rgenome.net/cas-analyzer/
CRISPResso2		https://crispresso.pinellolab.partners.org/
CRISPRAnalyzeR		http://crispr-analyzer.dkfz.de/
CRISPR-Sub		http://www.rgenome.net/crispr-sub
BE-Designer	Base editing	http://www.rgenome.net/be-designer/
DeepBaseEditor		http://deepcrispr.info/DeepBaseEditor
BE-Hive		https://www.crisprbehive.design/
BE-Analyzer		http://www.rgenome.net/be-analyzer/
PE-Designer	Prime editing	http://www.rgenome.net/pe-designer
PrimeDesign		https://drugthatgene.pinellolab.partners.org/
PE-Analyzer		http://www.rgenome.net/pe-analyzer/
DeepPE		http://deepcrispr.info/DeepPE/
pegFinder		http://pegfinder.sidichenlab.org

Guide Verification tools - IDT

☐ Checker analysis: Sequence1 — Custom checker Alt-R CRISPR-Cas9 gRNA

Sequence	On-target score	Off-target score
TGTCTCGCGAGCTCAGAGCG	22	89

Hide off-target details - | Show related products

+ ADD TO DESIGN SET

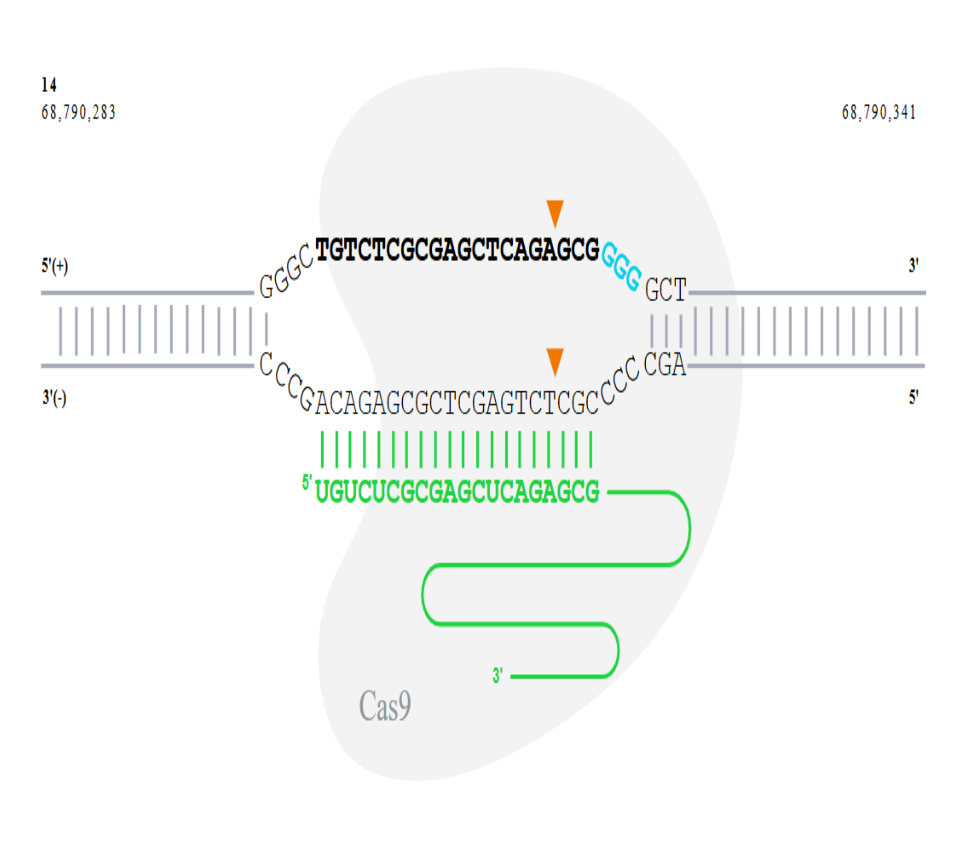
CREATE NEW DESIGN

Off-target results Demonstrates off-target hits to corresponding gRNA Design. Export to Excel | Export to CSV

Sequence	PAM	Score	#MM	Gene	Locus
TGTCTCGCGAGCTCAGAGCG	GGG	N/A		ZFP36L1	chr14:+68790299
TGTCACAGCAGCTCAGAGCG	CAG	21	4		chr18:-251793
TGTTTCATGAGCTCAGAGTG	TGG	31	4		chr1:-60293221
TGTCTCGC-AGCTCAGAGCA	TGG	36	2		chr17:-78526399
AGTCTCGCGAGCCCGGAGCG	TGG	37	3	NPRL3	chr16:-138651
TGTCTCG-GAGCTCAGGGCG	CAG	38	2	ARF3	chr12:+48937327
TCTGTCGCTGGCTCAGAGCG	GGG	43	4		chr2:-240820638
TTTCTCAAGAGCTCAGAGCT	AAG	46	4		chr5:-39232040
TGTCTCTCGAACTCGGAGCT	GGG	46	4	ZC3HC1	chr7:-130023680

https://eu.idtdna.com/site/order/designtool/index/CRISPR_SEQUENCE

Guide Verification tools - Synthego



OFF TARGET SEQUENCES

Filters

OFF TARGET SITE	MISMATCHES	CHROMOSOME	CUT SITE	PAM	GENE
AGTCTCGCGAGCCCGGAGCG	3	16	138,655	TGG	NPRL3
TGTCTCGTGAGGTTAGAGTG	4	11	34,667,128	AGG	
TGTCTGTCAAGCTCAGAGCA	4	7	44,384,737	TGG	NUDCD3
TGTCTCGGAAACTCAGAGCC	4	5	112,861,308	GGG	CTC-554D6.1
TGTCTCCCCAGCTCACAGCC	4	CHR_HG30_PATCH	179,903,182	TGG	
TGTCTCCCAAGCTCAGGGCC	4	10	93,070,733	TGG	
TGTCTCTGGAGCTCAGCGCA	4	4	111,753,901	TGG	
AGGCACGCGAGTTCAGAGCG	4	GL000205.2	56,922	TGG	

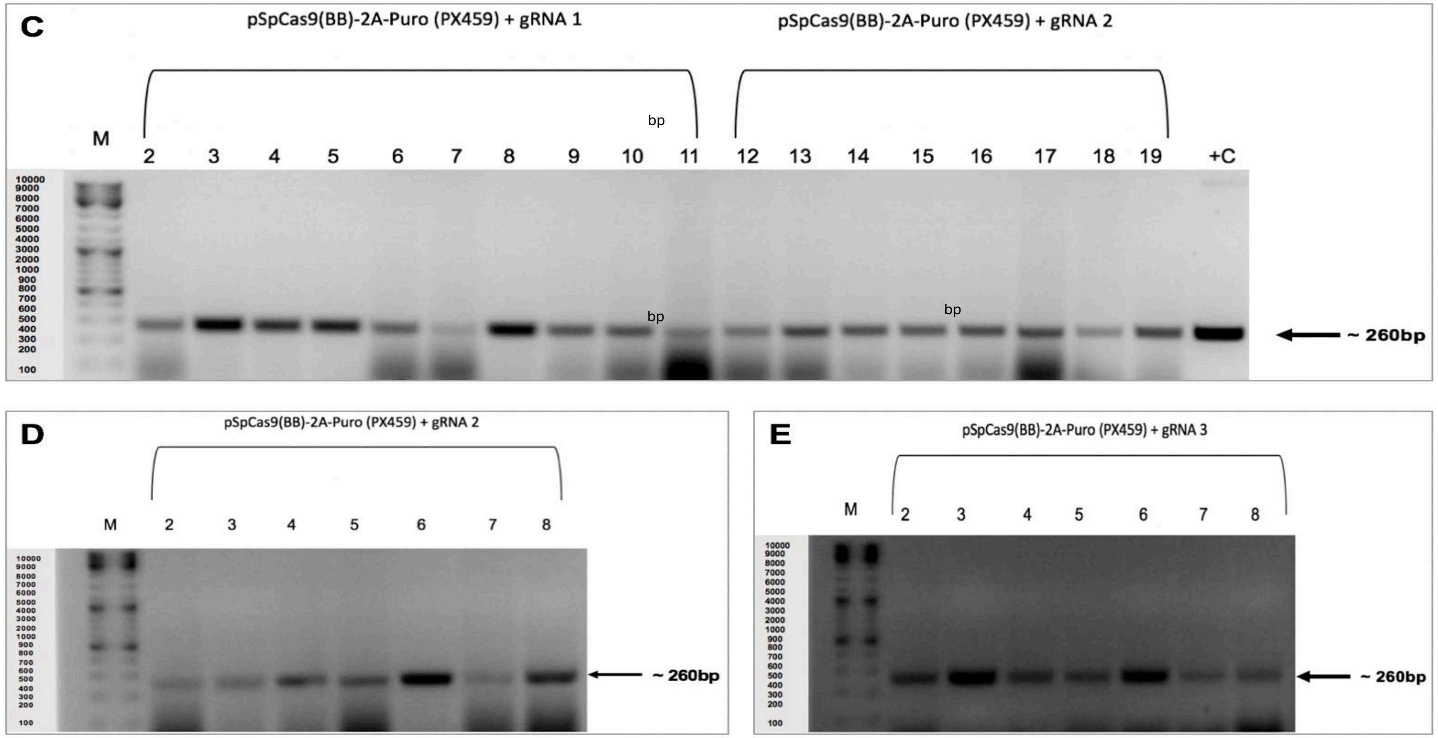
https://www.synthego.com/help/guide-verification-tool
https://www.frontiersin.org/journals/genome-editing/articles/10.3389/fgeed.2021.673022/full

Construction of the guide expression plasmids

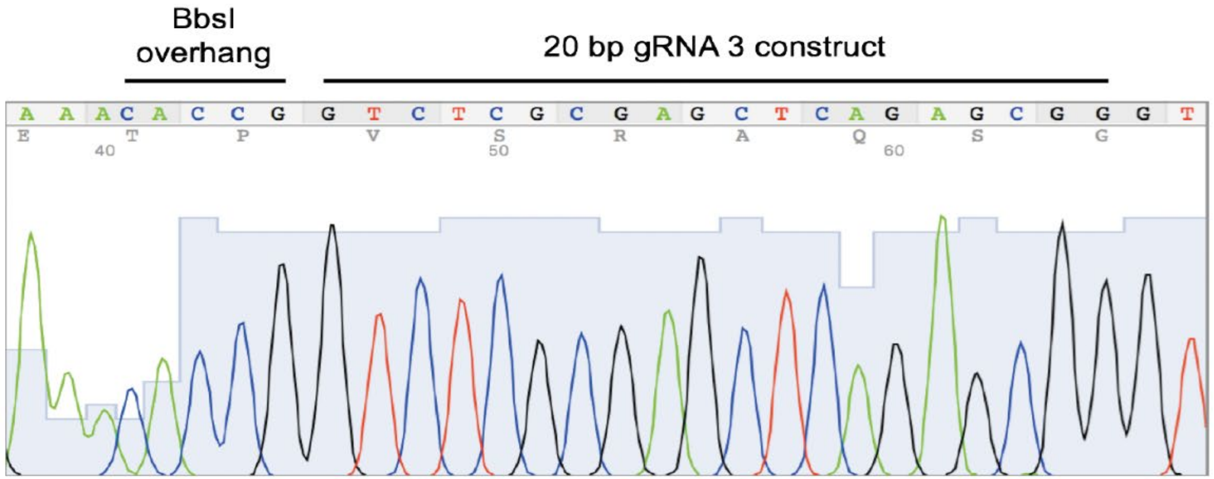
Stage 1

Choosing the Target → sgRNA design → Guide cloning into CRISPR vector → Sequence Verification

Bacterial Colony PCR



Plasmids sequence verification



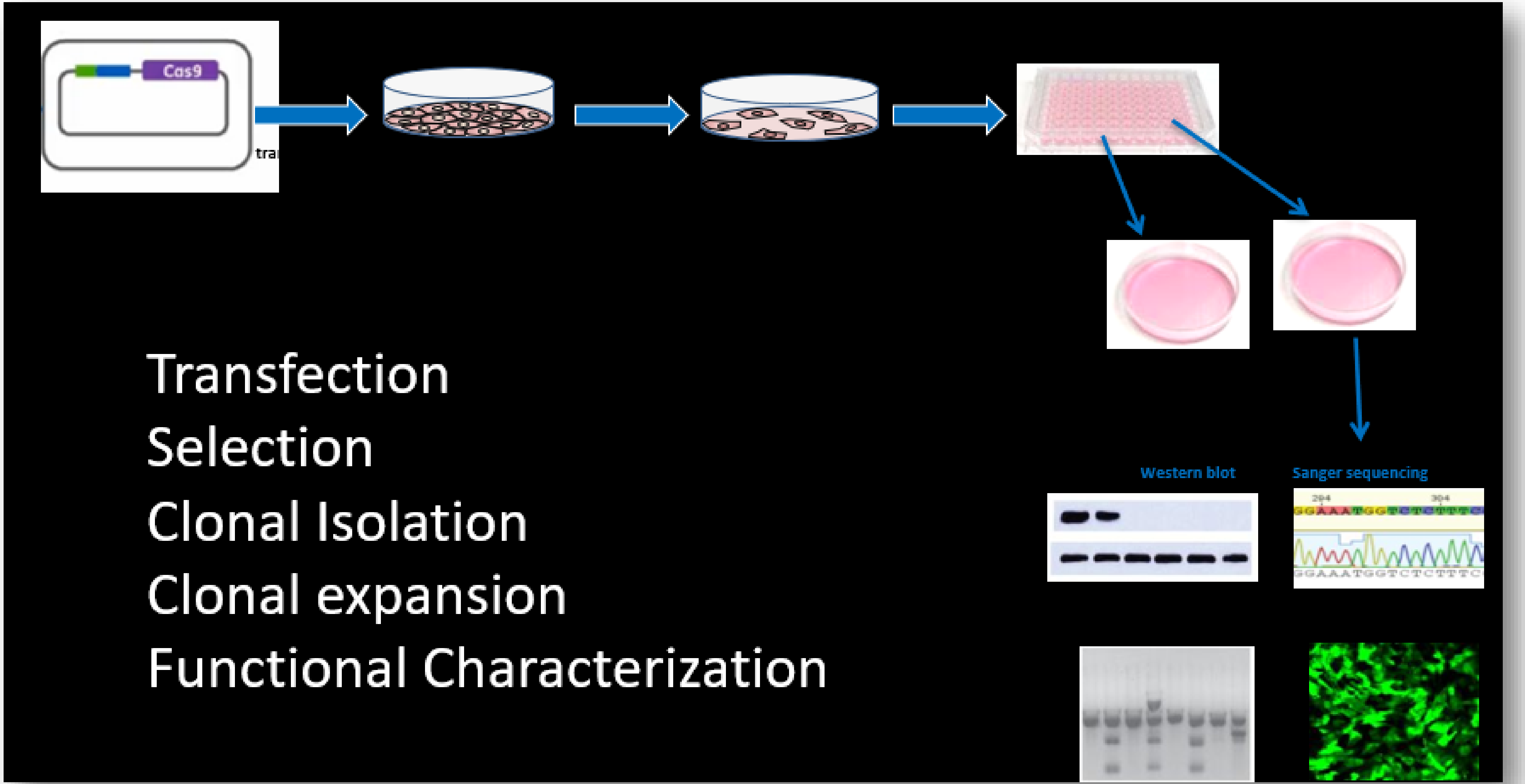
BbsI overhang

gRNA 3 insert

TTTATATATCTTGTGNAAAGGACGAAACACCGGTCTCGCGAGCTCAGAGCGGGTTTTA
GAGCTAGAAATAGCAAGTTAAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCA
CCGAGTCGGTGCTTTTTTGTTTTAGAGCTAGAAATAGCAAGTTAAAATAAGGCTAGTCC
GTTTTTAGCGCGTGCGCCAATTCTGCAGACAAATGGCTCTAGAGGTACCCGTTACATA
ACTTACGGTAAATGGCCCGCCTGGCTGACCGCCCAACGACCCCCGCCCATTGACGT
CAATAGTAACGCCAATAGGGACTTTCCATTGACGTCAATGGGTGGAGTATTTACGGTAA
ACTGCCCACTTGGCAGTACATCAAGTGTATCATATGCCAAGTACGCCCCCTATTGACG
TCAATGACGGTAAATGGCCCGCCTGGCATTGTGCCCAGTACATGACCTTATGGGACTT
TCCTACTTGGCAGTACATCTACGTATTAGTCATCGCTATTACCATGGTCGAGGTGAGCC
CCACGTTCTGCTTCACTCTCCCCATCTCCCCCCCCTCCCCACCCCCAATTTTGTATTT
ATTTATTTTTTAATTATTTTGTGCAGCGATGGGGGCGGGGGGGGGGGGGGGGTTTG

U6- Forward primer

Gene Editing workflow - Stage 2



Gene Editing Workflow- Stage 2& 3

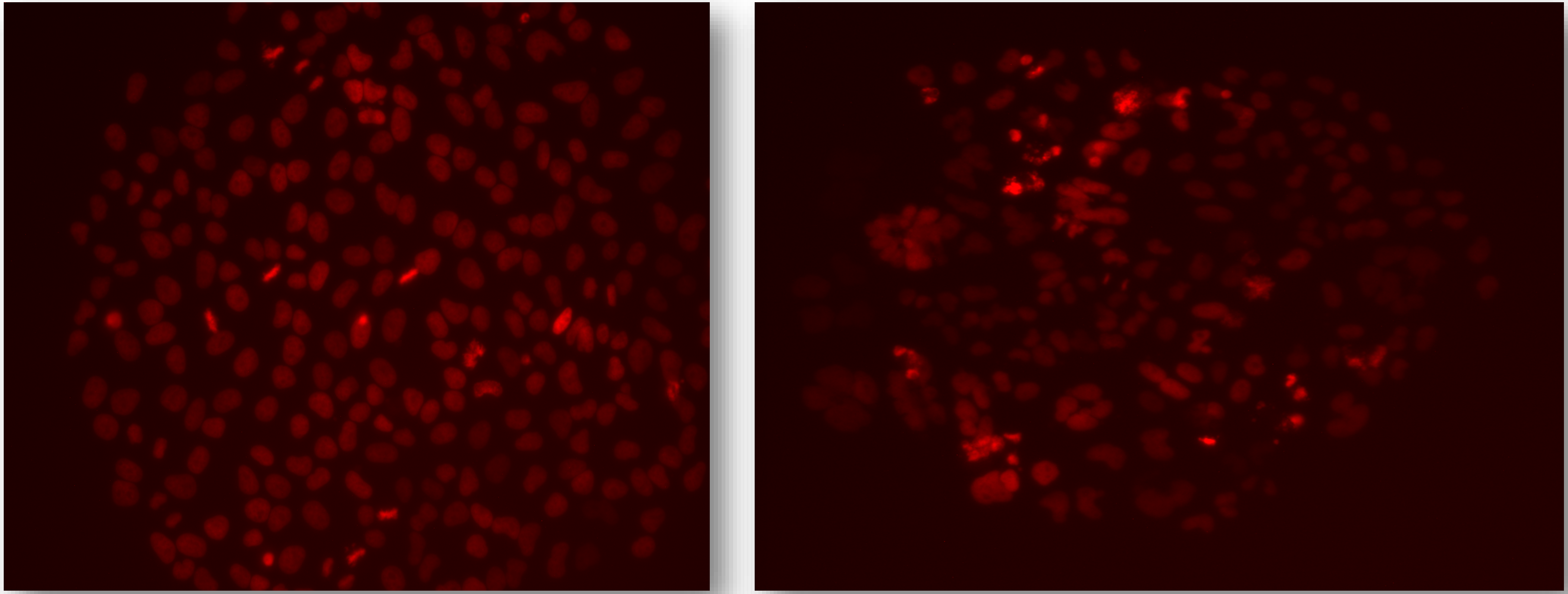
Stage2
~ 2 months

6. Transfect cells
7. Selection e.g. Puromycin
8. Clonal Isolation and expansion

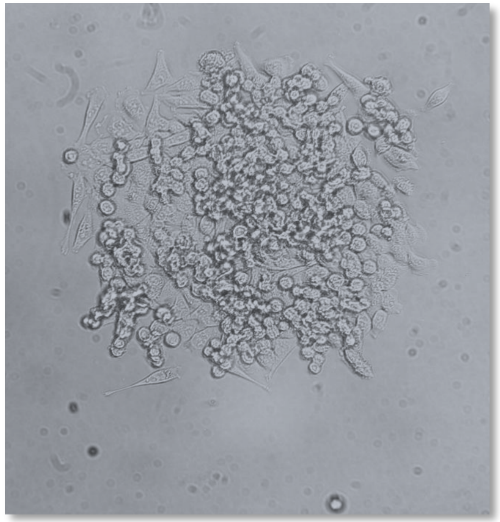
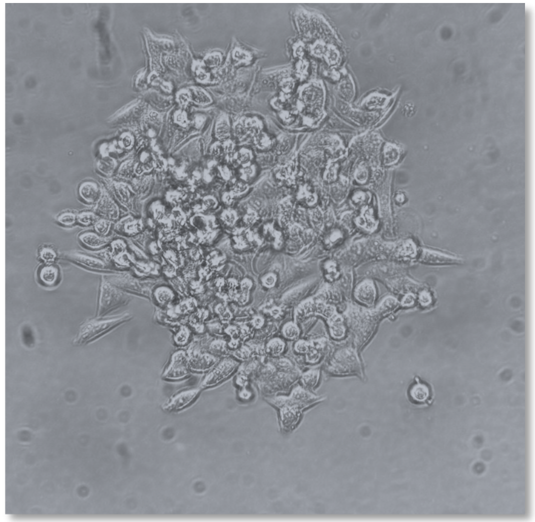
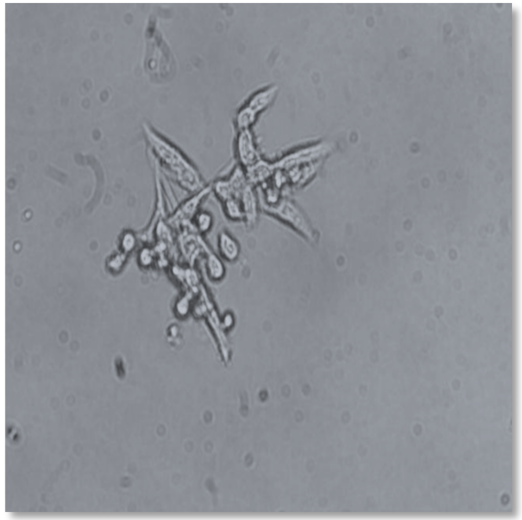
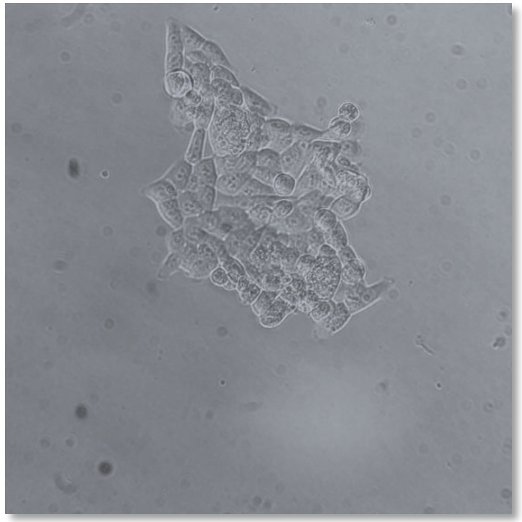
Stage 3

9. Functional characterization and phenotype and genotype analysis

Colony picking

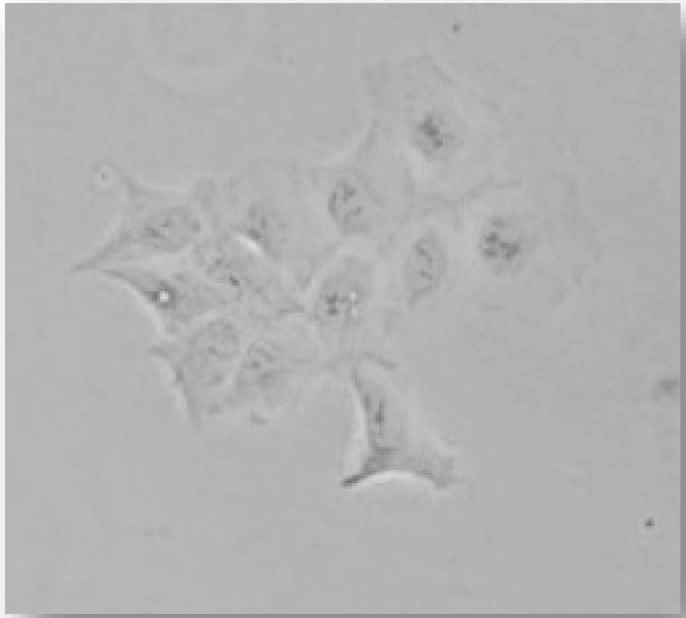


Colony picking

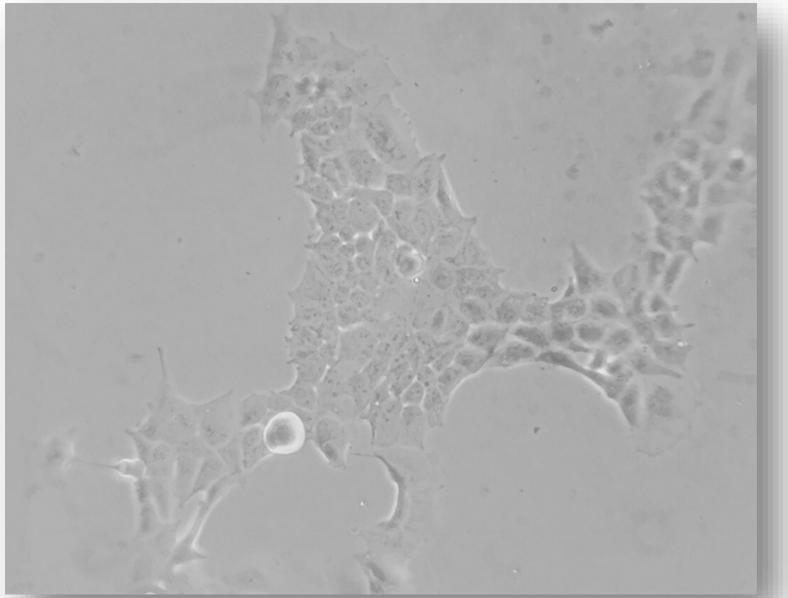
Clonal selection – observe early!

HEK293



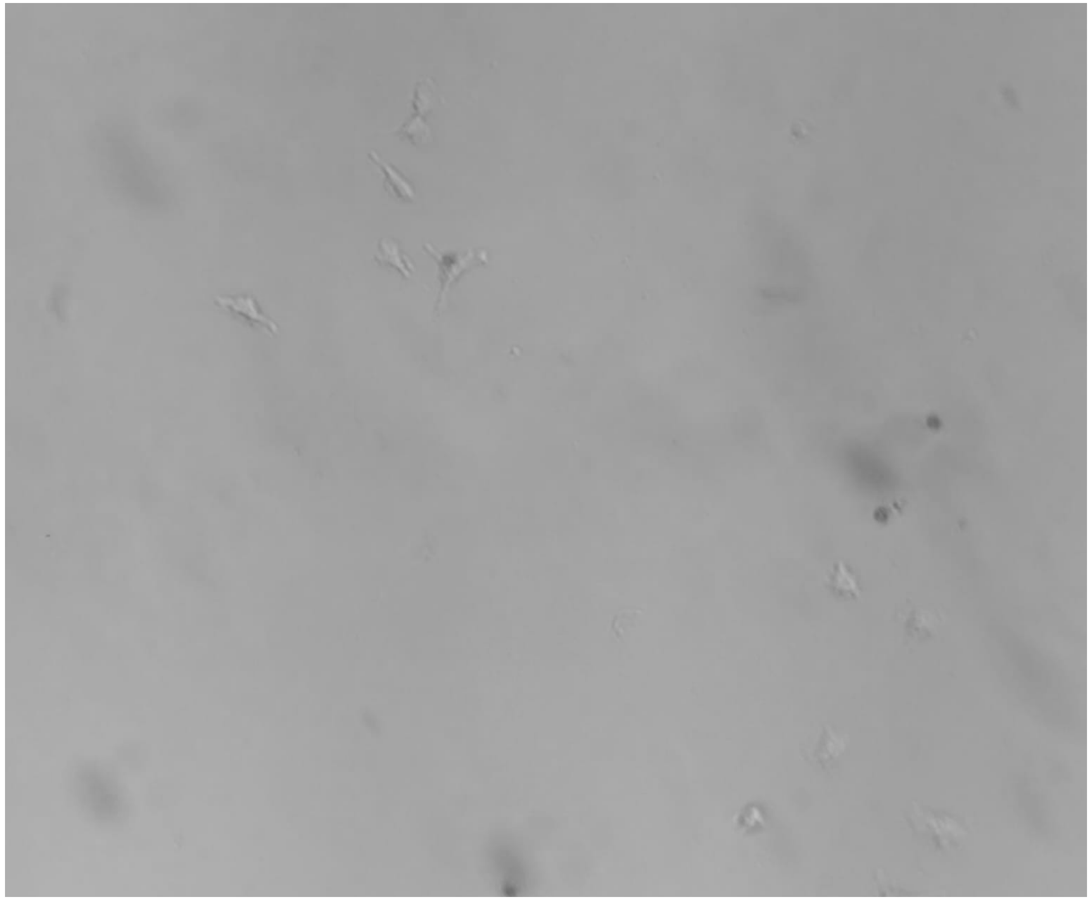
A clone of ~10cells

HEK293

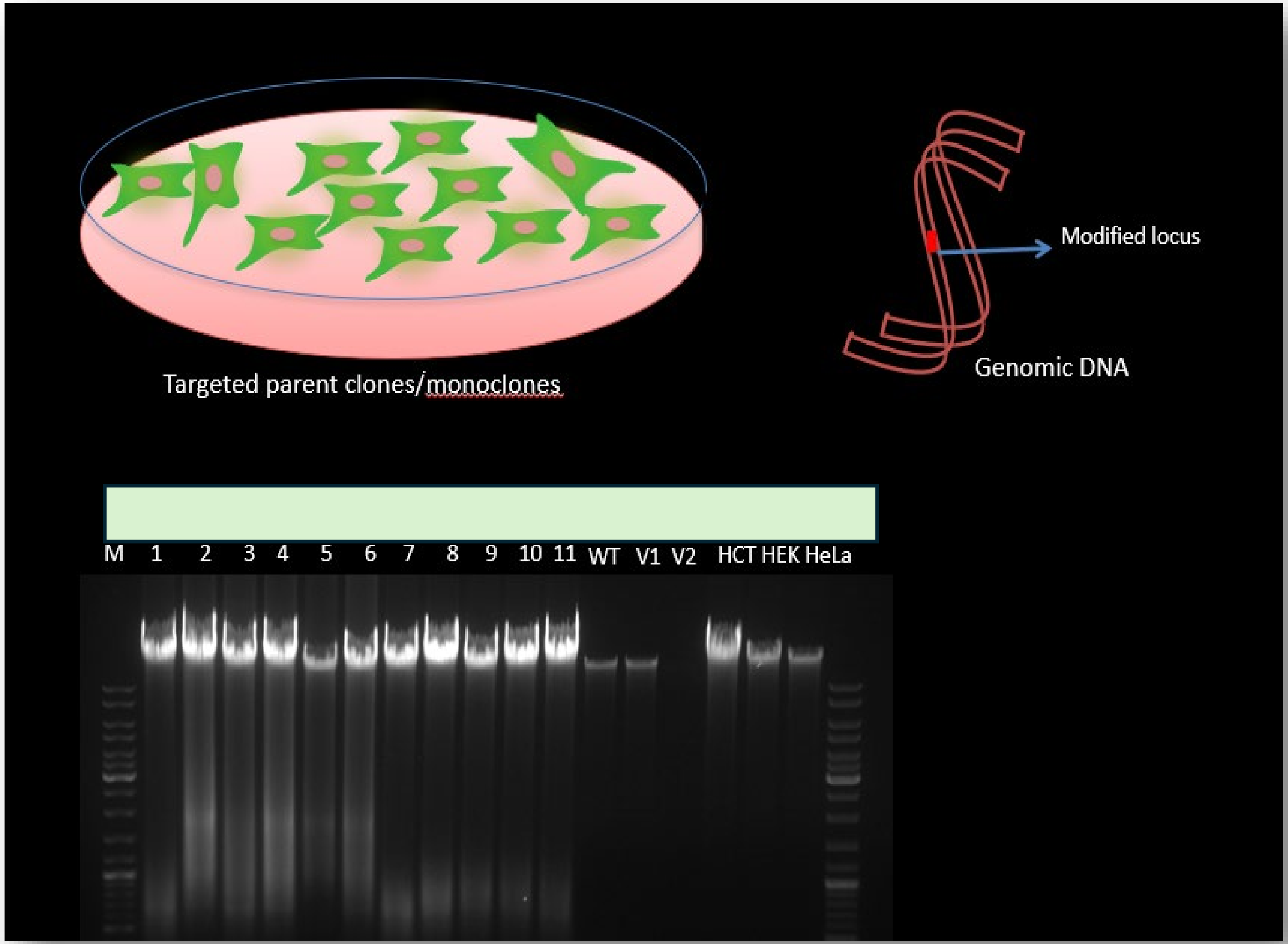


A clone of ~100cells

Antibiotic selection – not too early!



PCR genotyping

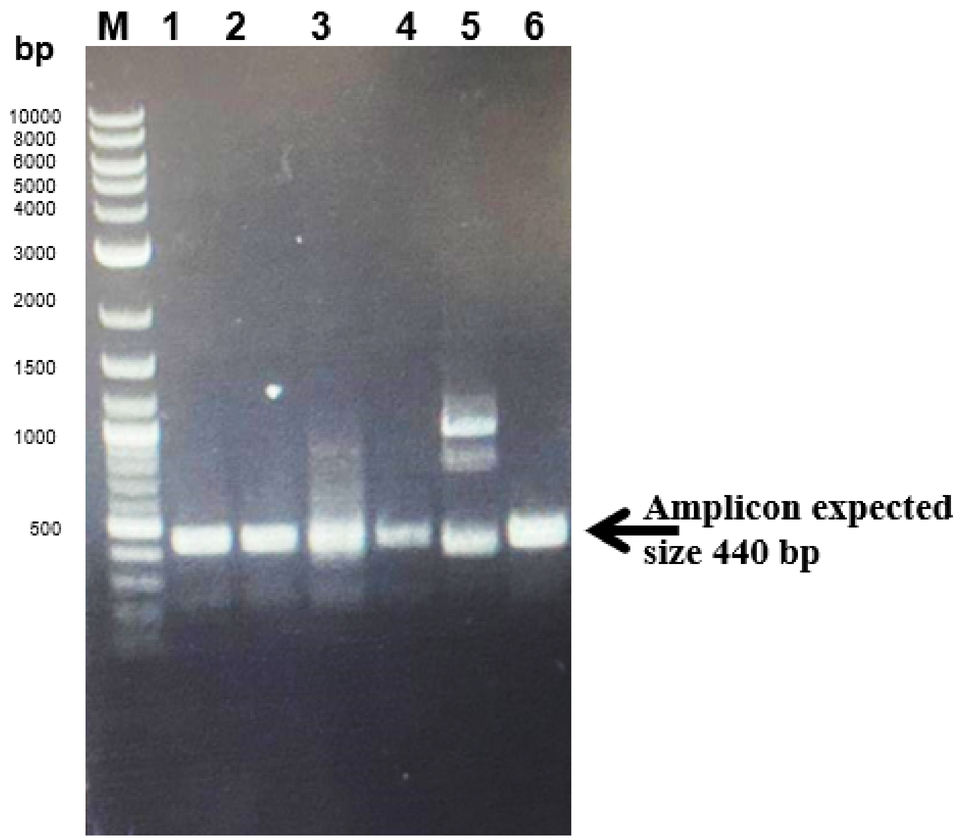


Design genotyping primers during the guide RNA design

https://www.genewiz.com/en-gb/public/services/next-generation-sequencing/amplicon-sequencing-services

https://www.genewiz.com/en-gb/public/services/sanger-sequencing/sanger-ez

PCR genotyping



ATCTCTGCTCCTATTGGCTGGATATTTCGTATTCCCCGAGCTCCTAA
AAACG gPCRFor
*AACCAATAGGAAGAGCGGAC*AGCGATCTCTAACGCGCAAGCGCATA
TCC**TTCTAGGTAGCGGGCAGTAG**CCGCTTCAGGGAGGGACGAA
GAGACCCAGCAACCCACAGAGTTGAGAAATTTGACTGGCATTCAA
GCTGTCCAATCAATAGCTGCCGCTGAAGGGTGGGG***CTGGATGGCG***
TAAGCTACAGCTGAAGGAAGAACGTGAGCACGAGGCACTGAGGT
GATTGGCTGAAGGCACTTCCGTTGAGCATCTAGACGTTTCCTTGGC
TCTTCTGGCGCCAAAATGTCGTTCGTGGCAGGGGTTATTCGGCGG
CTGGACGAGACAGTGGTGAACCGCATCGCGGCGGGGGAAGTTATC
CAGCGGCCAGCTAATGCTATCAAAGAGATGATTGAGAACTGGTAC
GGAGGGAGTCGAGCCG*GGCTCACTTAAGGGCTACGA* ***gPCRRev***
CTTAACGGGCCGCGTCACTCAATGGCGCGGACACGCCTCTTTGCC
CGGGCAGAGGCATGTACAGCGCATGCCCACAACGGCGGAGGCCG
CCGGGTTCCCTGACGTGCCAGTCAGGCCTTCTCCTTTTCCGCAGA
CCGTGT

Gene Editing Verification

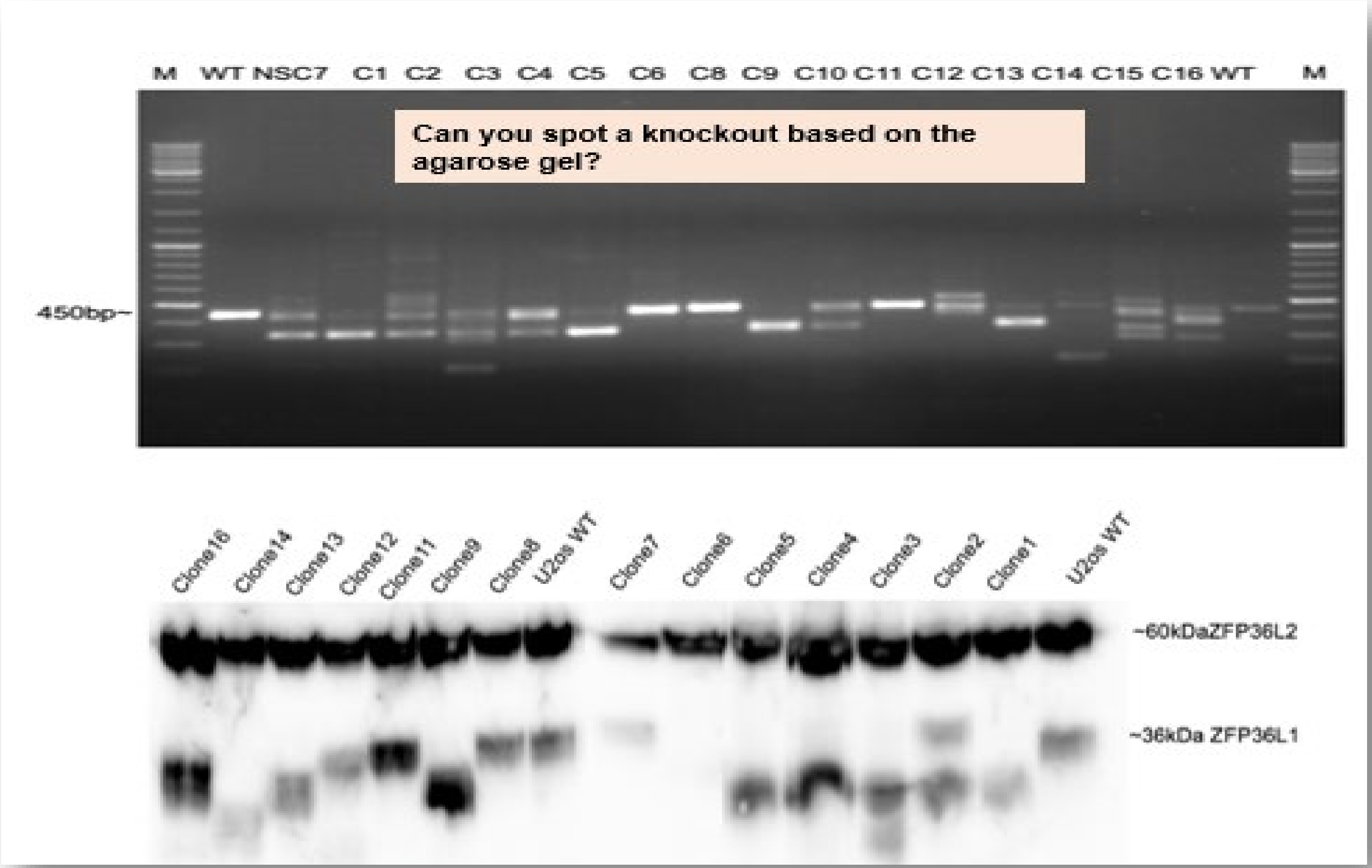
GENEWIZ®
From Azenta Life Sciences

rhAmpSeq™ Design Tool

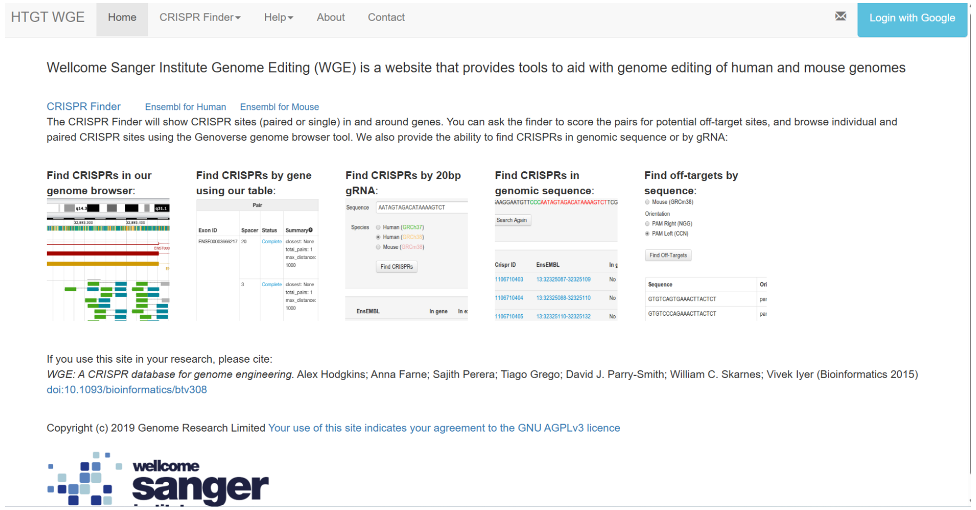
Verification of Gene Editing in Mixed pools and Monoclonal lines

in CRISPRESSO session by Khalid Akram

Gene Editing Verification




TIME TO
TAKE A
BREAK


HTGT WGE
Home
CRISPR Finder
Help
About
Contact
Login with Google
Wellcome Sanger Institute Genome Editing (WGE) is a website that provides tools to aid with genome editing of human and mouse genomes
CRISPR Finder
Ensembl for Human
Ensembl for Mouse
The CRISPR Finder will show CRISPR sites (paired or single) in and around genes. You can ask the finder to score the pairs for potential off-target sites, and browse individual and paired CRISPR sites using the Genoverse genome browser tool. We also provide the ability to find CRISPRs in genomic sequence or by gRNA:
Find CRISPRs in our genome browser:
Find CRISPRs by gene using our table:
Find CRISPRs by 20bp gRNA:
Find CRISPRs in genomic sequence:
Find off-targets by sequence:
If you use this site in your research, please cite:
WGE: A CRISPR database for genome engineering. Alex Hodgkins; Anna Farne; Sajith Perera; Tiago Grego; David J. Parry-Smith; William C. Skarnes; Vivek Iyer (Bioinformatics 2015)
doi:10.1093/bioinformatics/btv308
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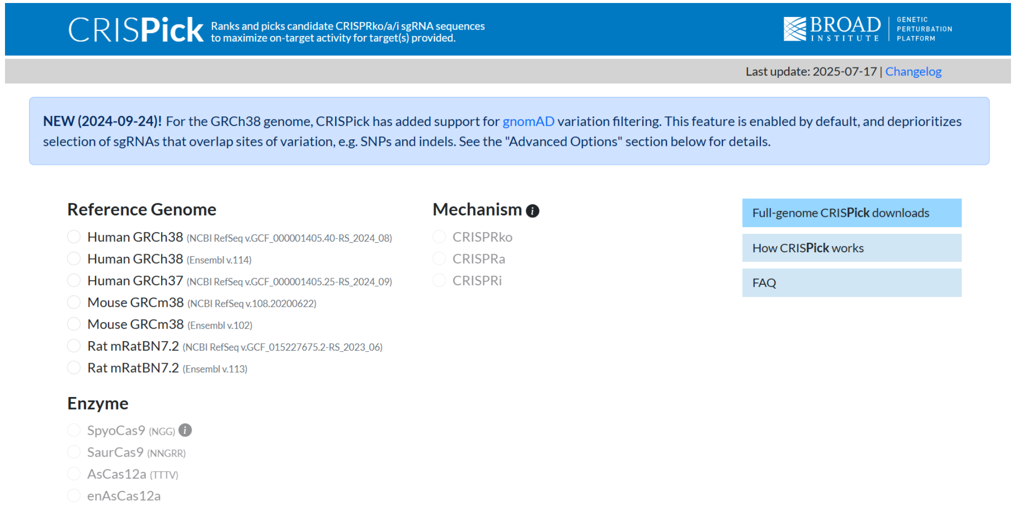
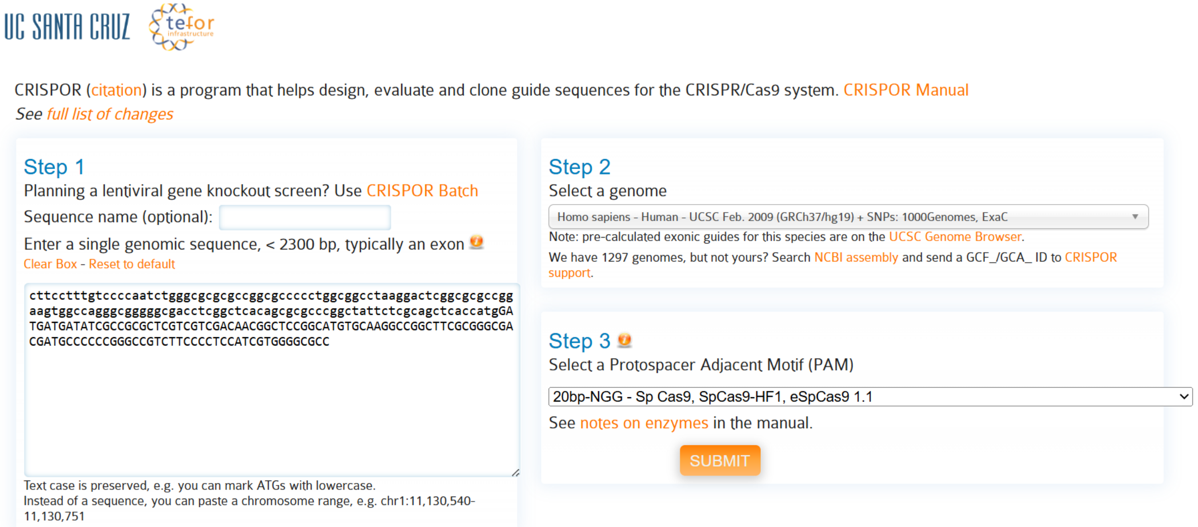
wellcome sanger


CRISPick
Ranks and picks candidate CRISPRko/a/i sgRNA sequences to maximize on-target activity for target(s) provided.
BROAD INSTITUTE
GENETIC PERTURBATION PLATFORM
Last update: 2025-07-17 | Changelog
NEW (2024-09-24)! For the GRCh38 genome, CRISPick has added support for gnomAD variation filtering. This feature is enabled by default, and deprioritizes selection of sgRNAs that overlap sites of variation, e.g. SNPs and indels. See the "Advanced Options" section below for details.
Reference Genome
Human GRCh38 (NCBI RefSeq v.GCF_000001405.40-RS_2024_08)
Human GRCh38 (Ensembl v.114)
Human GRCh37 (NCBI RefSeq v.GCF_000001405.25-RS_2024_09)
Mouse GRCm38 (NCBI RefSeq v.108.20200622)
Mouse GRCm38 (Ensembl v.102)
Rat mRatBN7.2 (NCBI RefSeq v.GCF_015227675.2-RS_2023_06)
Rat mRatBN7.2 (Ensembl v.113)
Mechanism
CRISPRko
CRISPRa
CRISPRi
Full-genome CRISPick downloads
How CRISPick works
FAQ
Enzyme
SpyoCas9 (NGG)
SaurCas9 (NNGRR)
AsCas12a (TTTV)
enAsCas12a


UC SANTA CRUZ
tefor infrastructure
CRISPOR (citation) is a program that helps design, evaluate and clone guide sequences for the CRISPR/Cas9 system. CRISPOR Manual
See full list of changes
Step 1
Planning a lentiviral gene knockout screen? Use CRISPOR Batch
Sequence name (optional):
Enter a single genomic sequence, < 2300 bp, typically an exon
Clear Box - Reset to default
cttcctttgtccccaatctgggcgcgcgccggcgccccctggcggcctaaggactcggcgcgccgg
aagtggccagggcgggggcgacctcggctcacagcgcgcccggctattctcgcagctcaccatgGA
TGATGATATCGCCGCGCTCGTCGTCGACAACGGCTCCGGCATGTGCAAGGCCGGCTTCGCGGGCGA
CGATGCCCCCCGGGCCGTCTTCCCCTCCATCGTGGGGCGCC
Text case is preserved, e.g. you can mark ATGs with lowercase.
Instead of a sequence, you can paste a chromosome range, e.g. chr1:11,130,540-11,130,751
Step 2
Select a genome
Homo sapiens - Human - UCSC Feb. 2009 (GRCh37/hg19) + SNPs: 1000Genomes, ExaC
Note: pre-calculated exonic guides for this species are on the UCSC Genome Browser.
We have 1297 genomes, but not yours? Search NCBI assembly and send a GCF_/GCA_ ID to CRISPOR support.
Step 3
Select a Protospacer Adjacent Motif (PAM)
20bp-NGG - Sp Cas9, SpCas9-HF1, eSpCas9 1.1
See notes on enzymes in the manual.
SUBMIT
Version 5.2 - Documentation - Contact us - Downloads/local installation - Citation - License

CRISPR Guide RNA Design Tool

The CRISPR Guide RNA design tool allows you to visualize, optimize, and annotate multiple gRNA sequences at a time. Get on and off-target scores in seconds to compare and optimize for higher activity and lower off-target effects. Save, organize and attach guides to relevant sequences and assemble gRNAs into plasmids — all within Benchling.

Sign up for free   Request a demo

Useful links

- https://www.sanger.ac.uk/tool/wge-crispr-design-tool/
- https://www.opentargets.org/
- https://cellmodelpassports.sanger.ac.uk/
- https://depmap.org/portal/achilles/
- https://sites.broadinstitute.org/ccle/
- https://depmap.org/portal/
- https://depmap.sanger.ac.uk/

All the best!


HOW DOES HE DO IT??
YEARS OF PRACTICE. MANY... MANY... YEARS...
ZZZZZ ...
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